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Wound therapy system with internal alternating orifice

Abstract

A wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, tubing fluidly communicable with the wound space, and a canister fluidly communicable with the tubing. The canister, the tubing, and the dressing define a sealed space that includes the wound space. The wound therapy system also includes a therapy unit coupled to the canister. The therapy unit includes a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and controllable between an open position and a closed position, and a control circuit. The control circuit is configured to control the valve to alternate between the open position and the closed position to allow airflow through the valve, receive measurements from the sensor, and determine a volume of the wound space based on the measurements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 16/363,928, filed on Mar. 25, 2019, which claims the benefit of priority to U.S. Provisional Application No. 62/802,034, filed on Feb. 6, 2019, all of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) The present disclosure relates generally to a wound therapy system, and more particularly to a wound therapy system that provides negative pressure wound therapy (NPWT). NPWT refers to the creation of negative pressure (relative to atmospheric pressure) at a wound to promote healing of the wound. In a wound therapy system configured to provide NPWT, a dressing is typically sealed over a wound bed and placed in fluid communication with a pump operable to draw a negative pressure at the wound bed (i.e., in a wound space between the wound bed and the dressing). Because the dressing is sealed over the wound bed, often for a period of multiple days, it may be difficult to ascertain and monitor the progress of wound healing. One way to determine an amount of wound healing is based on a change in the amount of volume between the wound bed and the dressing (i.e., as the wound heals into the volume to occupy/consume part of the volume). Accordingly, systems and methods for volume determination in a wound therapy system may be advantageous.

(2) In some cases, NPWT may be provided in coordination with instillation therapy and described as negative pressure and instillation wound therapy (NPIWT). Instillation therapy refers to the provision of instillation fluid (e.g., saline, antibiotic fluid) to the wound. One challenge in instillation therapy may be determining how much fluid to provide to the wound. It may be preferable to determine an amount of fluid to provide based on a size of the wound and/or a volume of available space adjacent the wound (i.e., between the dressing and the wound). Accordingly, systems and methods for volume determination in a wound therapy system may facilitate instillation therapy.

SUMMARY

(3) One implementation of the present disclosure is a wound therapy system. The wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, tubing coupled to the dressing and fluidly communicable with the wound space, and a canister fluidly communicable with the tubing. The canister, the tubing, and the dressing define a sealed space that includes the wound space. A therapy unit is coupled to the canister. The therapy unit includes a pneumatic pump fluidly communicable with the sealed space, a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and controllable between an open position and a closed position, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve to repeatedly alternate between the open position and the closed position to allow a controlled rate of airflow through the valve, receive measurements of the pressure in the sealed space from the sensor, and determine a volume of the wound space based on the measurements of the pressure.

(4) In some embodiments, the controlled rate of airflow is less than a restriction rate of a filter positioned between the valve and the canister.

(5) In some embodiments, valve includes a solenoid valve. The control circuit is configured to control the valve to repeatedly alternate between the open position and the closed position by providing a voltage pattern to the solenoid valve. The voltage pattern includes a step function repeatedly stepping between approximately zero voltage and a non-zero voltage. The voltage pattern may remain at the non-zero voltage for no more than a maximum continuous duration of approximately 500 milliseconds.

(6) In some embodiments, the voltage pattern includes a repeating pattern of approximately 400 milliseconds at a non-zero voltage, approximately 100 milliseconds at approximately zero voltage,

approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage. The voltage pattern includes a first set of two periods of the repeating pattern, approximately one second at approximately zero voltage, and a second set of two periods of the repeating pattern. The voltage pattern may cause the solenoid valve to alternate between the open position and the closed position with a period of approximately 500 milliseconds.

(7) In some embodiments, the control circuit is further configured to customize a customized wound therapy based on the volume of the wound space and control the therapy unit to provide the customized wound therapy. The customized wound therapy may include instillation therapy.

(8) In some embodiments, the control circuit is configured to customize the instillation therapy by determining an amount of instillation fluid to supply to the wound space based on the volume of the wound space. The wound therapy system may include instillation tubing coupled to the dressing and fluidly communicable with the wound space, a source of the instillation fluid fluidly communicable with the instillation tubing, and an instillation pump controllable by the control circuit to provide the amount of the instillation fluid from the source to the wound space.

(9) Another implementation of the present disclosure is a method of treating a wound. The method includes establishing a sealed space defined by a dressing, tubing, and a canister of a wound therapy system. The sealed space includes a wound space defined by the dressing and the wound. The method includes removing, with a pneumatic pump, air from the sealed space to establish a negative pressure in the sealed space and causing a solenoid valve to alternate between an open position and a closed position. The solenoid valve allows an airflow from a surrounding environment to the sealed space in the open position and prevents the airflow from the surrounding environment to the sealed space in the closed position. The method also includes measuring the pressure in the sealed space to generate pressure measurements, determining, based on the pressure measurements, a volume of the wound space, customizing a customized wound therapy based on the volume of the wound space, and providing the customized wound therapy to the wound.

(10) In some embodiments, customizing a customized wound therapy includes determining an amount of an instillation fluid to be supplied to the wound space based on the volume of the wound space. Providing the customized wound therapy to the wound includes controlling an instillation pump to supply the amount of the instillation fluid to the wound space.

(11) In some embodiments, causing the solenoid valve to alternate between the open position and the closed position provides a controlled rate of airflow from the surrounding environment to the sealed space. The controlled rate of airflow is less than a restriction rate of a filter positioned between the canister and the solenoid valve.

(12) In some embodiments, causing the solenoid valve to alternate between the open position and the closed position includes providing a voltage pattern to the solenoid valve. The voltage pattern may include a step function repeatedly stepping between approximately zero voltage and a non-zero voltage. The voltage pattern may include a repeating pattern of approximately 400 milliseconds at a non-zero voltage, approximately 100 milliseconds at approximately zero voltage, approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage.

(13) In some embodiments, the voltage pattern includes a first set of two periods of the repeating pattern, approximately one second at approximately zero voltage, and a second set of two periods of the repeating pattern. The non-zero voltage causes the solenoid valve to be in the open position. A positive pressure of approximately 5 mmHg is provided to the sealed space during each 400 milliseconds at the non-zero voltage.

(14) Another implementation of the present disclosure is a wound therapy system. The wound therapy system includes a dressing sealable over a wound and defining a wound space between the dressing and the wound, first tubing coupled to the dressing and fluidly communicable with the wound space, a canister fluidly communicable with the first tubing, wherein the canister, the first tubing, and the dressing define a sealed space that includes the wound space, a pneumatic pump

fluidly communicable with the sealed space, a sensor configured to measure a pressure in the sealed space, and a solenoid valve controllable between an open position and a closed position. The solenoid valve is configured to allow air to flow from a surrounding environment to the sealed space in the open position and prevent air from flowing from the surrounding environment to the sealed space in the closed position. The wound therapy system also includes instillation tubing coupled to the dressing and fluidly communicable with the wound space and a source of instillation fluid, an instillation pump coupled to the instillation tubing and controllable to supply an amount of the instillation fluid to the wound space, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space and provide a voltage pattern to the solenoid valve. The voltage pattern causes the solenoid valve to repeatedly alternate between the open position and the closed position. The control circuit is also configured to receive measurements of the pressure from the sensor, determine a volume of the wound space based on the measurements of the pressure, determine the amount of the instillation fluid based on the volume of the wound space, and control the instillation pump to supply the amount of the instillation fluid to the wound space.

(15) In some embodiments, causing the solenoid valve to alternate between the open position and the closed position allows a controlled rate of airflow through the solenoid valve from the surrounding environment to the sealed space.

(16) In some embodiments, the solenoid valve is positioned to allow the air to enter one or more outer lumens of the first tubing. In some embodiments, a filter is positioned between the solenoid valve and the one or more outer lumens. Causing the solenoid valve to alternate between the open position and the closed position allows a controlled rate of airflow through the solenoid valve from the surrounding environment to the channel, and the controlled rate is less than a restriction rate of the filter.

(17) In some embodiments, the instillation pump, the pneumatic pump, and the control circuit are housed within a therapy unit. In some embodiments, the solenoid valve is positioned within the therapy unit. In some embodiments, the solenoid valve is positioned outside the therapy unit and coupled to the first tubing.

(18) Another implementation of the present disclosure is a therapy unit. The therapy unit includes a pneumatic pump fluidly communicable with a sealed space, a sensor configured to measure a pressure in the sealed space, a valve positioned between the sealed space and a surrounding environment and controllable between an open position and a closed position, and a control circuit. The control circuit is configured to control the pneumatic pump to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve to repeatedly alternate between the open position and the closed position to allow a controlled rate of airflow through the valve, receive measurements of the pressure in the sealed space from the sensor, and determine a volume of the sealed space based on the measurements of the pressure and the controlled rate, and provide a customized wound therapy based on the volume of the sealed space.

(19) In some embodiments, the control circuit is configured to allow the controlled rate of airflow through the valve by controlling the valve to the open position for no longer than a maximum continuous duration of approximately 500 milliseconds.

(20) Another implementation of the present disclosure is a wound therapy system. The wound therapy system includes a pneumatic pump fluidly communicable with a canister and tubing comprising a first lumen and a second lumen. The first lumen is configured to facilitate the flow of fluid from a dressing to the canister and the second lumen configured to facilitate measurement of a pressure at the dressing. The wound therapy system also includes a sensor configured to measure a pressure in the second lumen, a valve positioned between a second lumen and a surrounding environment and controllable between an open position and a closed position, a filter positioned between the valve and the second lumen, and a cap removeably coupleable to the tubing. The cap provides fluid communication between the first lumen and the second lumen when the cap is

coupled to the tubing. The wound therapy system also includes a control circuit configured to, while the cap is coupled to the tubing, operate the pump to remove air from the canister, control the valve to the open position, receive measurements of the pressure in the second lumen from the sensor, and determine, based on the measurements of the pressure in the second lumen, a flow rate through the filter.

(21) In some embodiments, the control circuit is configured to, while the cap is removed from the tubing and the dressing is coupled to the tubing, determine a volume of a wound space based on the flow rate through the filter and additional measurements of the pressure from the sensor. In some embodiments, the control circuit is configured to provide a customized wound therapy based on the volume of the wound space.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 is a partial block diagram of a negative pressure wound therapy system including a therapy device coupled to a wound dressing via tubing, according to an exemplary embodiment.

(3) FIG. 2 is a block diagram illustrating the negative pressure wound therapy system of FIG. 1 in greater detail, according to an exemplary embodiment.

(4) FIG. 3 is a block diagram illustrating the negative pressure circuit, the removed fluid canister circuit and the wound site circuit of the negative pressure wound therapy system of FIG. 1 in greater detail, according to an exemplary embodiment.

(5) FIG. 4 is a block diagram illustrating a negative pressure wound therapy system, according to an exemplary embodiment.

(6) FIG. 5 is a flowchart of a method of using a negative pressure wound therapy system, according to an exemplary embodiment.

(7) FIG. 6A is a flowchart of method of instilling an initial quantity of fluid to a wound site using the negative pressure wound therapy system, according to an exemplary embodiment.

(8) FIG. 6B illustrates a negative pressure wound therapy system applied to a desired wound site to be treated, prior to the instillation of an initial volume of fluid to the wound site according to an exemplary embodiment.

(9) FIG. 6C illustrates the negative pressure wound therapy system of FIG. 6B following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(10) FIG. 6D illustrates the negative pressure wound therapy system of FIG. 6C during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 6C, according to an exemplary embodiment.

(11) FIG. 6E illustrates the negative pressure wound therapy system of FIG. 6B following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(12) FIG. 6F illustrates the negative pressure wound therapy system of FIG. 6E during venting of the negative pressure wound therapy system following the application of the second negative pressure as shown in FIG. 6E, according to an exemplary embodiment.

(13) FIG. 6G illustrates the instillation of fluid to the wound site using the wound therapy system of FIG. 6B, according to an exemplary embodiment.

(14) FIG. 7 illustrates a negative pressure wound therapy system applied to a wound site following an initial instillation of fluid to the wound site, according to an exemplary embodiment.

(15) FIG. 8A is a flowchart of method of instilling an additional quantity of fluid to a wound site using the negative pressure wound therapy system of FIG. 7, according to an exemplary embodiment.

(16) FIG. 8B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(17) FIG. 8C illustrates the negative pressure wound therapy system of FIG. 8B during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 8B, according to an exemplary embodiment.

(18) FIG. 8D illustrates the negative pressure wound therapy system of FIG. 7 following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(19) FIG. 8E illustrates the negative pressure wound therapy system of FIG. 8D during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 8D, according to an exemplary embodiment.

(20) FIG. 9A is a flowchart of method of instilling an additional quantity of fluid to a wound site to the negative pressure wound therapy system of FIG. 7, according to an exemplary embodiment.

(21) FIG. 9B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(22) FIG. 9C illustrates the negative pressure wound therapy system of FIG. 9B during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 9B, according to an exemplary embodiment.

(23) FIG. 9D illustrates the negative pressure wound therapy system of FIG. 7 following an application of a second negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(24) FIG. 9E illustrates the negative pressure wound therapy system of FIG. 9D during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 9D, according to an exemplary embodiment.

(25) FIG. 10A is a flowchart of a method of determining whether sufficient dead space is present in a negative pressure wound therapy system, according to an exemplary embodiment.

(26) FIG. 10B illustrates the negative pressure wound therapy system of FIG. 7 following an application of a first negative pressure to the negative pressure wound therapy system, according to an exemplary embodiment.

(27) FIG. 10C illustrates the negative pressure wound therapy system of FIG. 10B during venting of the negative pressure wound therapy system following the application of the first negative pressure as shown in FIG. 10B, according to an exemplary embodiment.

(28) FIG. 11 is a flowchart of a process for monitoring the healing progression of the wound site over time, according to an exemplary embodiment.

(29) FIG. 12 is a flowchart of a method of instilling an initial quantity of fluid to a wound site using the negative pressure wound therapy system, according to an exemplary embodiment.

(30) FIG. 13 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

(31) FIG. 14 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

(32) FIG. 15 illustrates a negative pressure wound therapy system including a tubeset module, according to an exemplary embodiment.

(33) FIG. 16A is a block diagram of a negative pressure wound therapy system including a tubeset

module, according to an exemplary embodiment.

(34) FIG. 16B illustrates the negative pressure wound therapy system comprising a tubeset module of FIG. 16A, according to an exemplary embodiment.

(35) FIG. 17 is a flowchart of a fully automated method of operating a tubeset module, according to an exemplary embodiment.

(36) FIG. 18 is a block diagram of a negative pressure and instillation wound therapy (NPIWT) system, according to an exemplary embodiment.

(37) FIG. 19 is a cross-sectional illustration of a solenoid valve of the NPIWT system of FIG. 18 in a closed position, according to an exemplary embodiment.

(38) FIG. 20 is a cross-sectional illustration of the solenoid valve of FIG. 19 in an open position, according to an exemplary embodiment.

(39) FIG. 21 is a cross-sectional illustration of tubing of the NPIWT system of FIG. 18, according to an exemplary embodiment.

(40) FIG. 22 is a flowchart of a process for managing blockages of the tubing of FIG. 21 using the solenoid valve of FIGS. 19-20, according to an exemplary embodiment.

(41) FIG. 23 is a flowchart of a process for volume determination by the NPIWT system of FIG. 18, according to an exemplary embodiment.

(42) FIG. 24 is a collection of graphs illustrating various aspects of the process of FIG. 23, according to an exemplary embodiment.

(43) FIG. 25 is a block diagram of the NPIWT system of FIG. 18 with a removable cap, according to an exemplary embodiment.

DETAILED DESCRIPTION OF THE FIGURES

Overview

(44) Referring generally to FIGS. 1-17, a wound therapy system is shown according to various exemplary embodiments. The wound therapy system may include a therapy device and a wound dressing. The therapy device may include an instillation fluid canister, a removed fluid canister, a valve, a pneumatic pump, an instillation pump, a tubeset module and a controller. The wound dressing can be applied to a patient's skin surrounding a wound. The therapy device can be configured to deliver instillation fluid to the wound and provide negative pressure wound therapy (NPWT) by maintaining the wound at negative pressure. Components of the wound therapy device, the wound dressing, and the wound site form a negative pressure circuit.

(45) The controller can be configured to operate the pneumatic pump, the instillation pump, the tubeset module and/or other controllable components of the therapy device. In some embodiments, the controller estimates the volume of the wound based on a comparison of observed dynamic pressure responses to negative pressure being applied to the entirety of the negative pressure circuit and negative pressure being applied to a selected portion of the negative pressure circuit. Based on the comparison of the observed dynamic responses, the controller may be configured to determine a quantity of instillation fluid to be delivered to the wound site.

(46) The tubeset module comprises one or more elements that are actuatable, controllable or which may otherwise be engaged by the controller, with the selective communication of the controller with the tubeset module being configured to allow the controller to, among other functions, effectuate and monitor various dynamic pressure responses in all of and/or in parts of the negative pressure circuit as needed to estimate the volume of the wound, determine a quantity of instillation fluid to be delivered to the wound site and/or perform any other number of functions that may be related to the use of the NPWT system 100.

(47) According to some embodiments, the volume relative to the wound site determined by the controller may relate to the dead space at the wound site (i.e. the available space within a drape layer applied about the wound site into which instillation fluid may be delivered). In some such embodiments, the controller may be configured to determine a quantity of instillation fluid to be delivered to the wound site based on a predetermined percentage of the calculated dead space

volume at the wound site (e.g., 20%, 50%, 80%, etc.). The controller can then operate the tubeset module and instillation pump to deliver the determined volume of instillation fluid to the wound. By basing the quantity of instillation fluid to be delivered to the wound site on a calculated volume of the dead space at the wound site, the negative pressure system may be configured to provide for more efficient and more precise delivery of instillation fluid, which may reduce the risk of leakage resulting from over-delivery of instillation fluid and the risk of ineffective wound site treatment resulting from under-delivery of instillation fluid.

(48) In some embodiments, the controller may additionally, or alternatively, measure and monitor volumes relative to the wound site at a plurality of times during wound treatment, with the controller determining healing progression of the wound site based on changes in the measured volume relative to the wound site over the course of NPWT treatment. By monitoring the healing progression of the wound site, the controller may be configured to alert a user if the healing of the wound site is not progressing as intended or expected. These and other features of the wound therapy system are described in detail below.

Wound Therapy System

(49) Referring now to FIG. 1, a negative pressure wound therapy (NPWT) system **100** is shown according to an exemplary embodiment. NPWT system **100** is shown to include a therapy device **102** fluidly connected to a wound dressing **112** via tubing **108** and **110**. As will be described in more detail below, according to various embodiments a tubeset module **300** may be operably connected to the tubing **108** and/or **110**.

(50) According to various embodiments, a wound dressing **112** may be placed on or within the wound site **114** and adhered or sealed to a patient's skin **116** surrounding a wound site **114** using drape layer **117**. Several examples of wound dressings **112** which can be used in combination with NPWT system **100** are described in detail in U.S. Pat. No. 7,651,484 granted Jan. 26, 2010, U.S. Pat. No. 8,394,081 granted Mar. 12, 2013, and U.S. patent application No. 14/087,418 filed Nov. 22, 2013. The entire disclosure of each of these patents and patent applications is incorporated by reference herein.

(51) As illustrated by the block diagram of FIG. 2, in general the therapy device **102** includes a pneumatic pump **120**, an instillation pump **122**, a filter **128**, and a controller **118**. Pneumatic pump **120** can be fluidly coupled to removed fluid canister **106** (e.g., via conduit **136**) and can be configured to draw a vacuum within removed fluid canister **106** by pumping air out of removed fluid canister **106**. In some embodiments, pneumatic pump **120** is configured to operate in both a forward direction and a reverse direction. For example, pneumatic pump **120** can operate in the forward direction to pump air out of removed fluid canister **106** and decrease the pressure within removed fluid canister **106**. Pneumatic pump **120** can operate in the reverse direction to pump air into removed fluid canister **106** and increase the pressure within removed fluid canister **106**. Pneumatic pump **120** can be controlled by controller **118**, described in greater detail below.

(52) Therapy device **102** can be configured to provide negative pressure wound therapy by reducing the pressure at wound site **114**. Therapy device **102** can draw a vacuum at wound site **114** (relative to atmospheric pressure) by removing wound exudate, air, and other fluids from wound site **114**. Wound exudate may include fluid that filters from a patient's circulatory system into lesions or areas of inflammation. For example, wound exudate may include water and dissolved solutes such as blood, plasma proteins, white blood cells, platelets, and red blood cells. Other fluids **121** removed from wound site **114** may include instillation fluid **105** previously delivered to wound site **114**. Instillation fluid **105** can include, for example, a cleansing fluid, a prescribed fluid, a medicated fluid, an antibiotic fluid, or any other type of fluid which can be delivered to wound site **114** during wound treatment. Instillation fluid **105** may be held in an instillation fluid canister **104** and controllably dispensed to wound site **114** via tubing **108**. In some embodiments, instillation fluid canister **104** is detachable from therapy device **102** to allow removed fluid canister **106** to be refilled and replaced as needed.

(53) Instillation pump **122** can be fluidly coupled to instillation fluid canister **104** via upstream instillation tubing **108a** and fluidly coupled to wound dressing **112** via downstream instillation tubing **108b**. Instillation pump **122** can be operated to deliver instillation fluid **105** to wound dressing **112** and wound site **114** by pumping instillation fluid **105** through upstream instillation tubing **108a** and downstream instillation tubing **108b**. Instillation pump **122** can be controlled by controller **118**, described in greater detail below. According to some embodiments, an instillation tubing valve **109** valve configured to allow for flow only in the direction from the instillation fluid canister **104** to the wound site **114** (e.g. via a one-way valve or a valve configured to be selectively switched by a user and/or by the controller **118** to a closed position prior to the application of negative pressure to the wound site **114**) may generally be provided at a location along a portion of the downstream instillation tubing **108b**. As will be described in more detail below, according to various embodiments, the instillation tubing valve **109** may be provided as part of the tubeset module **300**.

(54) Filter **128** can be positioned between removed fluid canister **106** and pneumatic pump **120** (e.g., along conduit **136**) such that the air pumped out of removed fluid canister **106** passes through filter **128**. Filter **128** can be configured to prevent liquid or solid particles from entering conduit **136** and reaching pneumatic pump **120**. Filter **128** may include, for example, a bacterial filter that is hydrophobic and/or lipophilic such that aqueous and/or oily liquids will bead on the surface of filter **128**. Pneumatic pump **120** can be configured to provide sufficient airflow through filter **128** that the pressure drop across filter **128** is not substantial (e.g., such that the pressure drop will not substantially interfere with the application of negative pressure to wound site **114** from therapy device **102**).

(55) Removed fluid canister **106** may be a component of therapy device **102** configured to collect wound exudate and other fluids **121** removed from wound site **114**. In some embodiments, removed fluid canister **106** is detachable from therapy device **102** to allow removed fluid canister **106** to be emptied and replaced as needed. A lower portion of removed fluid canister **106** may be filled with wound exudate and other fluids **107** removed from wound site **114**, whereas an upper portion of removed fluid canister **106** may be filled with air. Therapy device **102** can be configured to draw a vacuum within removed fluid canister **106** by pumping air out of removed fluid canister **106**. The reduced pressure within removed fluid canister **106** can be translated to wound dressing **112** and wound site **114** via tubing **110**.

(56) As shown in FIG. 1, disposed along tubing **110** at a location between the removed fluid canister **106** and the wound site **114** is a tubing valve **111** configured to selectively permit and prevent fluid flow between the removed fluid canister **106** and the wound site **114**. The tubing valve **111** may be defined by any number of different structures (e.g. spring-biased; duck-bill; clamp; check-valve, etc.) configured to allow for the selective control of fluids through the tubing **110**, and may include valves that are configured to be selectively opened and/or closed by a user, in response to a sensed stimulus (e.g. a predetermined threshold pressure), or by the controller **118**. As will be described in more detail below, according to various embodiments, the tubing valve **111** may be provided as part of the tubeset module **300**.

(57) Referring to the block diagram of FIG. 3, when the tubing valve **111** is in an open, flow configuration, removed fluid canister **106**, tubing **110** (i.e. both upstream tubing portion **110a** and downstream tubing portion **110b**), conduit **136** extending between pneumatic pump **120** and removed fluid canister **106**, the portion of downstream instillation tubing **108b** extending between the drape layer **117** and instillation tubing valve **109**, and wound site **114** are fluidly connected to define a negative pressure circuit **200**. Referring further to FIG. 3, when the tubing valve **111** is in a closed, no-flow configuration, the removed fluid canister **106**, conduit **136** and an upstream tubing portion **110a** of the tubing **110** extending between the removed fluid canister **106** and the tubing valve **111** define a removed fluid canister circuit **202** that is fluidly isolated from a wound site circuit **204** defined by the wound site **114**, a downstream tubing portion **110b** of tubing **110**

extending between the tubing valve **111**, a portion of downstream instillation tubing **108b** extending between the drape layer **117** and instillation tubing valve **109**, and the wound site **114**. As will be discussed in more detail below, the volumes of the tubing **110**, conduit **136**, and portion of downstream instillation tubing **108b** extending between the drape layer **117** and instillation tubing valve **109** define known volumes which can be easily subtracted from or otherwise factored into calculations of volume(s) relative to the wound site **114**.

(58) Referring again to FIG. **1**, according to some embodiments, also provided along and operably fluidly connected to tubing **110** at a location upstream of tubing valve **111** and downstream of removed fluid canister **106** is a calibrated leak system **113** defined by a vent **113a** formed through an outer wall of the tubing **110**, the vent **113a** being selectively closeable by a vent valve **113b**. Also forming a part of calibrated leak system **113** may be a flow detector **113c** configured to measure airflow through the vent **113a**. As will be described in more detail below, calibrated leak system **113** is configured to selectively control and measure airflow between tubing **110** and the ambient environment surrounding therapy device **102**. According to various embodiments, calibrated leak system **113** can be selectively opened to allow airflow into tubing **110** at a known, predetermined rate. As will be described in more detail below, according to various embodiments, calibrated leak system **113** may be provided as part of the tubeset module **300**.

(59) As will be described in more detail below, when both the vent valve **113b** and the tubing valve **111** are closed, operation of the pneumatic pump **120** may be configured to draw a vacuum in only the removed fluid canister circuit **202** portion of the negative pressure circuit **200** (such as, e.g., illustrated in FIG. **6E**). When the vent valve **113b** is closed and the tubing valve **111** is open, operation of the pneumatic pump **120** may be configured to draw a vacuum in the entirety of the negative pressure circuit **200** (such as, e.g., illustrated in FIG. **6C**). When the vent valve **113b** is open and the tubing valve **111** is closed, airflow from the environment around therapy device **102** may enter through the vent **113a** of the calibrated leak system **113** and fill the vacuum within the removed fluid canister circuit **202** (such as, e.g., illustrated in FIG. **6F**). As illustrated, e.g., by FIG. **6D**, when both the vent valve **113b** and the tubing valve **111** are open, airflow from the environment around therapy device **102** may enter through the vent **113a** of the calibrated leak system **113** and fill the vacuum within the entirety of the negative pressure circuit **200**. As will be understood, according to various embodiments, the opening and/or closing of the vent valve **113b** and/or tubing valve **111** may be effectuated manually or automatically, e.g., using tubeset module **300**.

(60) Although the calibrated leak system **113** has been disclosed as being positioned in-line with a portion of the tubing **110** extending between the wound site **114** and the removed fluid canister **106**, according to some embodiments, such as, e.g., illustrated in FIG. **4**, the calibrated leak system **113** may be instead formed in-line with conduit **136**. The operation of the calibrated leak system **113** of the embodiment of FIG. **4** is similar to the operation of the calibrated leak system **113** illustrated in FIG. **1**, with the calibrated leak system **113** of FIG. **4** being configured to provide a path through which air from the ambient environment may flow into and fill portions or the entirety of the negative pressure circuit **200** following the creation of a vacuum within a portion or entirety of the negative pressure circuit **200**. As will be understood, according to various embodiments, any of the methods or systems illustrated or disclosed herein which incorporate a calibrated leak system **113** embodiment as illustrated in FIG. **1** may be modified with a calibrated leak system **113** embodiment as illustrated in FIG. **4**.

(61) As illustrated by the block diagram of FIG. **2**, according to various embodiments, the controller **118** may be configured to operate various components of therapy device **102**. In particular, as will be described in more detail below, according to various embodiments, the controller **118** may be configured to control the various components of the NPWT system **100** to execute one or more volume determination procedures via which, e.g., a quantity of instillation fluid **105** to be delivered to the wound site **114** may be determine, the healing progression of the

wound site **114** may be tracked, etc. According to various embodiments, the controller **118** may be configured such that these procedures may be performed with minimal user intervention and/or input.

(62) According to various embodiments, therapy device **102** may include a variety of sensors. For example, in some embodiments, therapy device **102** may include pressure sensor **115a** and/or **115b** located in-line in the upstream tubing portion **110a** and/or downstream tubing portion **110b**, which are configured to measure pressure at the removed fluid canister **106** and/or wound site **114**.

Pressure measurements recorded by pressure sensor(s) **115a** and/or **115b** can be communicated to controller **118**. According to various embodiments, controller **118** may use the pressure measurements from pressure sensor(s) **115a** and/or **115b** as inputs to various pressure testing operations and control operations performed by controller **118**. As will be described in more detail below, according to various embodiments, the pressure sensor(s) **115a** and/or **115b** may be provided as part of the tubeset module **300**.

(63) In some embodiments, therapy device **102** includes a user interface **126**. User interface **126** may include one or more buttons, dials, sliders, keys, or other input devices configured to receive input from a user. User interface **126** may also include one or more display devices (e.g., LEDs, LCD displays, etc.), speakers, tactile feedback devices, or other output devices configured to provide information to a user. User interface **126** can also display alerts generated by controller **118**. For example, controller **118** can generate a “no canister” alert if removed fluid canister **106** is not detected.

(64) In some embodiments, therapy device **102** includes a data communications interface **124** (e.g., a USB port, a wireless transceiver, etc.) configured to receive and transmit data.

(65) Communications interface **124** may include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications external systems or devices. In various embodiments, the communications may be direct (e.g., local wired or wireless communications) or via a communications network (e.g., a WAN, the Internet, a cellular network, etc.). For example, communications interface **124** can include a USB port or an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. In another example, communications interface **124** can include a Wi-Fi transceiver for communicating via a wireless communications network or cellular or mobile phone communications transceivers.

Methods of Use

(66) Referring to FIG. 5, a flowchart of a method **500** of using NPWT system **100** according to an exemplary embodiment is shown. As will be discussed in more detail with reference to FIGS. 6A-6G, initial set up of the NPWT system **100** and a delivery of an initial amount of instillation fluid **105** to a wound site **114** being treated by the NPWT system **100** occurs at step **502**.

(67) As shown at step **504**, according to various embodiments, it may be desirable to deliver additional instillation fluid **105** to the wound site **114** following the instillation of an initial amount of instillation fluid **105** to the wound site **114**. As will be understood, the determination at step **504** of when and if additional instillation fluid **105** is to be delivered to the wound site **114** may be based on any number of various factors, including e.g. elapsed time from a prior instillation; type of wound site **114**; desired course of wound site **114** treatment; sensed conditions related to the wound site **114**, etc., and may be decided automatically by the controller **118**, or may be based on user input.

(68) If it is determined at step **504** that additional fluid is to be delivered, at step **506** the dead space **119** at the wound site **114** is determined according to any of the methods as will be described below. According to various embodiments (described in more detail below), at step **506**, the controller **118** may be configured to determine the dead space **103** at the wound site **114** prior to such delivery of additional instillation fluid **105**, irrespective of: whether the quantity of instillation fluid **105** previously instilled to the wound site **114** is known; the presence of non-absorbed

instillation fluid **105** and/or wound exudate in the space defined between the wound site **114** and the drape layer **117**; whether the volume of any contents **107** in the removed fluid canister **106**, the volume of the removed fluid canister **106** itself, and/or the volume of any contents **107** previously emptied from the removed fluid canister **106** are known; whether the removed fluid canister **106** has been replaced with a different-sized removed fluid canister **106** during the course of the NPWT treatment; changes to the shape/size/volume of the wound site **114**; etc.

(69) At step **508**, the quantity of additional instillation fluid **105** to be delivered to the wound site **114** is calculated. According to various embodiments, the quantity of additional instillation fluid **105** delivered to the wound site **114** may be based on the volume of the dead space determined at step **506**. For example, in some embodiments, the controller **118** may calculate the volume of instillation fluid **105** to be delivered to wound site **114** by multiplying the volume of dead space determined at step **506** by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid **105** delivered to the wound site **114** does not exceed the available space within the drape layer **117** (i.e. dead space), thereby minimizing the risk of inadvertent leakage from the wound dressing **112**/drape layer **117**. In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8.

(70) In addition to being used to calculate instillation fluid **105** volumes, in some embodiments, the NPWT system **100** may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site **114** over time. Accordingly, in some embodiments, method **500** may optionally include the step **510** of estimating wound site **114** volume, and using the estimated volume to track healing progress of the wound site **114**, discussed in more detail with reference to FIG. **11** below.

(71) In some embodiments, it may be desired to remove instillation fluid **105** previously instilled to a wound site **114** from the wound site **114** at some time following the delivery of the instillation fluid **105** to the wound site **114**. Accordingly, it may be advantageous to confirm, prior to instilling instillation fluid **105** to the wound site **114**, that the dead space in the removed fluid canister **106** will be sufficient to receive the removed instillation fluid **105** and/or any additional fluid **121** (e.g. wound exudate) from the wound site **114** prior to delivering the additional instillation fluid **105** to the wound site **114**. As such, method **500** may optionally include step **512** at which the volume of additional instillation fluid **105** calculated at step **508** is compared to the dead space of the removed fluid canister **106** (measured, e.g., during the determination of dead space at the wound site **114** at step **506**), with an alarm being presented to the user at step **514** if the instillation fluid **105** to be delivered exceeds the dead space of the removed fluid canister **106**. If the instillation fluid **105** to be delivered does not exceed the dead space of the removed fluid canister **106** (or if step **512** is not included as part of method **500**), the calculated instillation fluid **105** is delivered to the wound site **114**, with some or all of steps **504**, **506**, **508**, **510**, **512**, **514**, **516** being repeated any number of additional times over the course of the NPWT treatment.

(72) Referring to FIG. **6A** a flowchart detailing the steps of a method **600** for an initial set up of NPWT system **100** and for delivery of an initial amount of instillation fluid **105** to a wound site **114** entailed in step **502** of the method **500** of FIG. **5** is shown according to one embodiment. At step **602**, a NPWT system **100** (such as, e.g., illustrated in FIG. **1**) is provided, with the drape layer **117** and wound dressing **112** being positioned at the desired wound site **114** to be treated, as shown, e.g., in FIG. **6B**.

(73) Once the set-up of the NPWT system **100** at step **502** is complete, the determination of the dead space **119** available at the wound site **114** into which instillation fluid **105** may be delivered may begin at step **604** with the controller **118** operating the pneumatic pump **120** to establish a first desired negative pressure within the entirety of the negative pressure circuit **200**, such as, e.g., illustrated in FIG. **6C**.

(74) In embodiments in which the tubing valve **111** comprises a normally-closed pressure sensitive

valve that is openable in response to an applied, predetermined threshold negative pressure, the first desired negative pressure generated by the controller **118** at step **604** may be equal to or greater than the predetermined threshold pressure required to open the tubing valve **111**, so as to ensure that the vacuum applied by the pneumatic pump **120** is applied across the entirety of the negative pressure circuit **200**. In some embodiments, the threshold pressure required to open the tubing valve **111** may be a pressure of approximately negative 125 mmHg, with the controller **118** being configured to apply at step **604** a first negative pressure that is equal to or greater than negative 125 mmHg.

(75) Alternatively, in embodiments in which the opening/closing of the tubing valve **111** is controlled manually or in direct response to a signal from the controller **118** (using, e.g., a tubeset module **300** as described below), the negative pressure delivered at step **604** may generally include any desired range of negative pressures, with step **604** including verification by the user and/or controller that the tubing valve **111** is in an open, flow orientation prior to the negative pressure being applied by the pneumatic pump **120**. As illustrated, e.g., in FIG. 6C, according to various embodiments, the instillation tubing valve **109** and the vent valve **113b** may be configured to be set to closed configurations during the application of negative pressure to the negative pressure circuit **200**.

(76) As illustrated by FIG. 6D, at step **606**, following the attainment of the desired first negative pressure within the negative pressure circuit **200** (as, e.g., measured and reported to the controller **118** by pressure sensor **115a** and/or pressure sensor **115b**), the operation of the pneumatic pump **120** is stopped, and the vent valve **113b** is opened to allow air from the ambient environment surrounding the therapy device **102** to flow through the vent **113a** and into the negative pressure circuit **200**. According to various embodiments, the opening of the vent valve **113b** at step **606** may be effectuated manually by a user or in response to instructions from the controller **118** being transmitted to the tubeset module **300**. In yet other embodiments, the calibrated leak system **113** may be formed without a vent valve **113b** (i.e. the vent **113a** defines a constant leak within the tubing **110**), such that air from the ambient environment surround the therapy device **102** will flow into the negative pressure circuit **200** without requiring any user and/or controller **118** intervention.

(77) As air from the ambient environment flows in to the negative pressure circuit **200**, parameters related to the flow of air through the vent **113a** into the negative pressure circuit **200** are monitored (e.g. via flow detector **113c**, pressure sensor **115a**, pressure sensor **115b**, etc.), with the measured parameters subsequently being used by the controller **118** at step **612** to determine the volume of the negative pressure circuit **200**. According to various embodiments, the parameters related to the flow of air into the negative pressure circuit **200** may include, e.g., the rate of flow of air into the negative pressure circuit **200** (as measured, e.g., by flow detector **113c**), the duration of time required for pressure within the negative pressure circuit **200** to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent **113a** and/or following operation of the pump **120** being ceased, the changing pressure (as, e.g., measured by pressure sensor **115a** and/or pressure sensor **115b**) within the negative pressure circuit **200** as the pressure increases from the negative pressure applied at step **604** to the predetermined pressure, etc.

(78) Once the pressure within the negative pressure circuit **200** has increased to a desired pressure and the measurement of the desired parameters has been completed by the controller **118**, the controller **118** may be configured operate pneumatic pump **120** to establish a second desired negative pressure within the removed fluid canister circuit **202** portion of the negative pressure circuit **200** at step **608**, such as, e.g., illustrated in FIG. 6E. In embodiments in which the tubing valve **111** comprises a normally-closed pressure sensitive valve that is openable in response to an applied, predetermined threshold negative pressure, the second desired negative pressure generated by the controller **118** at step **608** may be less than the predetermined threshold pressure required to open the tubing valve **111**, so as to ensure that the vacuum applied by the pneumatic pump **120** at step **608** is applied across only the removed fluid canister circuit **202** portion of the negative

pressure circuit **200**. For example, in some embodiments, the threshold negative pressure required to open the tubing valve **111** may be approximately negative 125 mmHg, with the controller **118** being configured to apply a negative pressure at step **608** that is less than negative 125 mmHg, such as, e.g., a pressure of approximately negative 50 mmHg.

(79) Alternatively, in embodiments in which the opening/closing of the tubing valve **111** is controlled manually or in direct response to a signal from the controller **118**, the negative pressure delivered at step **608** may generally include any desired range of negative pressures, with step **608** including verification by the user and/or controller that the tubing valve **111** is in a closed, no-flow orientation prior to the negative pressure being applied by the pneumatic pump **120**. As will be understood, in such embodiments, the second negative pressure applied by the controller **118** at step **608** to the removed fluid canister circuit **202** may include a pressure that is equal to or different from the negative pressure that is applied by the controller **118** at step **604** to the negative pressure circuit **200**. As illustrated, e.g., in FIG. 6E, according to various embodiments, the instillation tubing valve **109** and the vent valve **113b** may be configured to be set to closed configurations (either manually or automatically, e.g., using tubeset module **300**) during the application of negative pressure to the removed fluid canister circuit **202** at step **608**.

(80) As illustrated by FIG. 6F, at step **610**, following the attainment of the desired second negative pressure within the removed fluid canister circuit **202** (as, e.g., measured and reported to the controller **118** by pressure sensor **115a** and/or pressure sensor **115b**), the operation of the pneumatic pump **120** is stopped, and air from the ambient environment surrounding the therapy device **102** is allowed to flow through the vent **113a** and into the removed fluid canister circuit **202**. As air from the ambient environment flows into the removed fluid canister circuit **202**, parameters related to the flow of air through the vent **113a** and into the removed fluid canister circuit **202** are monitored, with the measured parameters subsequently being used by the controller **118** to calculate the volume of the removed fluid canister circuit **202** at step **612**. According to various embodiments, the parameters related to the flow of air into removed fluid canister circuit **202** may include, e.g., the rate of flow of air into the removed fluid canister circuit **202** (as measured, e.g., by flow detector **113c**), the duration of time required for pressure within the removed fluid canister circuit **202** to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent **113a** and/or ceasing operation of the pump **120** at step **610**, the pressure (as, e.g., measured by pressure sensor **115a** and/or pressure sensor **115b**) within the removed fluid canister circuit **202** as the pressure increases from the negative pressure applied at step **608** to the predetermined pressure; etc.

(81) At step **612**, the controller **118** may be configured to determine the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** based on the parameters measured at steps **606** and **610**. According to some embodiments, the controller **118** may base these volume calculations on stored relationships between various measured parameter values and corresponding volumes. These relationships between measured parameter measurements and corresponding volumes that are stored by the controller **118** may include various functions, models, lookup table, etc., and may be based on pre-existing information input and stored by the controller **118**, or on information obtained and processed by the controller **118** during an optional, initial training procedure conducted by the controller **118** prior to the use of the NPWT system **100** to treat wound site **114** (e.g. prior to the initiation of method **500**; as part of the initial setup and initial instillation of instillation fluid of step **502**; etc.). One non-limiting examples of embodiments of training procedures by which such relationships may be generated by the controller **118** are outlined in related, U.S. Provisional Application 62/650,132, filed Apr. 17, 2018 and titled WOUND THERAPY SYSTEM WITH WOUND VOLUME ESTIMATION, the entire disclosure of which is incorporated by reference herein.

(82) Using the determined volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200**, the controller **118** may determine the volume of the dead space **119** at the

wound site **114** (i.e. the portion of the interior space defined between the wound site **114** and the lower surface of the drape layer **117** that is not occupied by the wound dressing **112** and/or any instillation fluid **105**/other fluid) by subtracting the volume of the removed fluid canister circuit **202** from the volume of the negative pressure circuit **200**. According to various embodiments, the determination of the volume of the dead space **119** at the wound site **114** at step **614** may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** to account for/factor in the known volumes of the downstream tubing portion **110b** and the portion of the downstream instillation tubing **108b** extending between the drape layer **117** and the instillation tubing valve **109** into the determination of the volume of the dead space **119** at the wound site **114**.

(83) At step **614**, an initial quantity of instillation fluid **105** that is to be delivered to the wound site **114** is calculated. According to various embodiments, the calculated initial quantity of instillation fluid **105** that is delivered to the wound site **114** may be based on the volume of the dead space **119** calculated by the controller **118** at step **612**. For example, in some embodiments, the controller **118** may calculate the initial volume of instillation fluid **105** to be delivered to the wound site **114** by multiplying the volume of dead space **119** calculated at step **612** by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid **105** delivered to the wound site **114** does not exceed the available space within the drape layer **117** (thereby minimizing inadvertent leakage from the wound dressing **112**/drape layer **117**). In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8. However, it is contemplated that the fluid instillation factor can have any value in various alternative embodiments.

(84) As noted previously with reference to step **510**, in addition to being used to calculate a quantity of instillation fluid **105** to be delivered during any stage of treatment using NPWT system **100** and under any number of different conditions (e.g. allowing for the calculation of additional instillation fluid **105** to be delivered at step **516** even if the removed fluid canister **106** has been emptied, or entirely replaced with a different sized removed fluid canister **106** during the course of treatment), in some embodiments the NPWT system **100** may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site **114** over time. Accordingly, in some embodiments, at step **616**, an initial baseline wound site **114** volume estimate may optionally be determined (via, e.g., a method as described with regards to FIG. **11** below) and stored by the controller **118**, which may be used as a reference point against which future wound site **114** volume estimates may be compared to track healing progression of the wound site **114**.

(85) For reasons similar to those described with reference to step **512** of the method **500** of FIG. **5**, according to some embodiments, at step **618** the amount of initial instillation fluid **105** that is to be delivered calculated at step **614** may be compared to a determined dead space **103** of the removed fluid canister **106** to determine whether the dead space within the removed fluid canister **106** will be sufficient to collect any fluids **121** from the wound site **114** (including non-absorbed instillation fluid **105**) following the delivery of instillation fluid **105** at step **516**. As will be understood, in embodiments in which the NPWT system **100** has not been operated prior to the use of the NPWT system **100** at step **602**, the volume of the removed fluid canister **106** should be empty, such that the dead space **103** of the removed fluid container **106** should be equal to the volume of the removed fluid canister **106**. If the volume of the removed fluid canister **106** is not known and/or if removed fluid **107** is present in the removed fluid canister **106** at step **602**, the dead space **103** of the removed fluid container may be calculated by subtracting the known volumes of conduit **136** and the upstream tubing portion **110a** from the volume of the removed fluid canister circuit **202** determined at step **614**. Similar to step **514**, at step **620** an alarm may be presented to a user if the initial volume of instillation fluid **105** to be delivered calculated at step **614** exceeds the dead space **103** of the removed fluid canister **106**. Otherwise, if the volume of the initial instillation fluid **105** to be delivered does not exceed the dead space **103** of the removed fluid canister **106**, the

calculated instillation fluid **105** is delivered to the wound site **114** at step **622**, as shown, e.g., in FIG. **6F**.

(86) Referring to FIG. **7**, a NPWT system **100** according to one embodiment is shown at a point in time subsequent to a decision to instill additional instillation fluid **105** to the wound site **114** at step **504** of the method **500** of FIG. **5**, but prior to the determination of wound dead space at the wound site at step **506**. As shown in FIG. **7**, at the time immediately preceding the determination of dead space at the wound site **114** at step **506**, a quantity of fluid **121** (e.g. non-absorbed instillation fluid **105** from a prior instillation, wound exudate, etc.) may be present in the space between the drape layer **117** and the wound site **114**, with the remaining space between the drape layer **117** and the wound site **114** defining an initial dead space **119a**. As also shown in FIG. **7**, according to some embodiments, an initial quantity of removed fluid **107** may be present in the removed fluid canister **106** at the time immediately preceding the start of step **506**, with the remaining volume of the removed fluid canister **106** being defined by an initial dead space **103a**. As will be understood, according to some embodiments, no fluid may be present at either the wound site **114** and/or in the removed fluid canister **106** at the time immediately preceding step **506**, in which embodiments the quantities of each of the fluid **121** in the wound space and the removed fluid **107** in the removed fluid canister **106** would be equal to zero.

(87) As noted above, a quantity of fluid **121** may be present at the wound site **114** immediately prior to the initiation of step **506**. According to some embodiments, it may not be desired and/or required to remove fluid **121** from the wound site (e.g. non-absorbed instillation fluid **105** from prior instillations, wound exudate, etc.) prior to the delivery of additional instillation fluid **105** to the wound site **114** at step **516** of the method **500** of FIG. **5**. Accordingly, in some embodiments of method **500**, the additional instillation fluid **105** instilled to the wound site at step **516** may be delivered to the wound site **114** irrespective of any fluid **121** that may be present at the wound site **114**.

(88) Referring to FIGS. **8A-8E**, one embodiment of a method **800** of determining an amount of dead space at a wound site **114** which may be used at step **506** of the method **500** of FIG. **5** in embodiments in which fluid **121** from the wound site **114** is not removed from the wound site **114** prior to instilling additional instillation fluid **105** is illustrated. In particular, according to the method **800** of FIGS. **8A-8E**, as no fluid **121** is displaced from the wound site **114** during the method **800** (i.e. step **506**), the final dead space into which the additional instillation fluid **121** will be instilled will be the same initial dead space **119a** at the wound site that is present immediately prior to the initiation of step **506** (i.e. the dead space **119a** shown in FIG. **7**).

(89) As shown by the flowchart in FIG. **8A**, the method **800** of determining dead space is substantially the same as the method **600** of calculating the dead space **119** upon initial instillation of instillation fluid **105** to the wound site **114** at step **502** (which is discussed in more detail with reference to FIGS. **6A-6G**). In particular, similar to steps **604** and **606**, the method **800** of FIG. **8A** also includes steps **802** and **804** (shown, e.g., in FIGS. **8B** and **8C**, respectively) during which negative pressure is applied to and removed from the negative pressure circuit **200**. Similar to steps **608** and **610** of the method **600** of FIG. **6A**, the method **800** of FIG. **8** also includes steps **806** and **808** (shown, e.g., in FIGS. **8D** and **8E**, respectively) during which negative pressure is applied to and removed from the removed fluid canister circuit **202**. Also similar to the method **600** of FIG. **6A**, in the method **800** of FIGS. **8A-8E**, the application and subsequent removal of negative pressure to the negative pressure circuit **200** of steps **802** and **804** may be performed either prior to or after the application and subsequent removal of negative pressure to the removed fluid canister circuit **202** of steps **806** and **808**.

(90) As noted above, the method **800** of FIGS. **8A-8E** may be performed in substantially the same manner as the method **600** described with references to FIG. **6A** above. However, whereas, as described above with reference to the method of FIGS. **6A-6E**, according to various embodiments, any range of negative pressures may generally be applied to the negative pressure circuit **200** at

step **604** of method **600**, the negative pressure applied to the negative pressure circuit **200** at step **802** of the method **800** must be limited to negative pressures that will not result in the fluid **121** at the wound site **114** being displaced into the removed fluid canister **106**.

(91) Following the completion of step **808**, the controller **118** may be configured to calculate the volume of the dead space **119a** at the wound site **114** (which corresponds to the maximum volume of additional instillation fluid **105** that may be delivered to wound site **114**) at step **508** of method **500** of FIG. 5. More specifically at step **508**, after calculating the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** based on the parameters measured at steps **804** and **808** (in a manner similar to that described with reference to step **612** of the method **600** of FIGS. **6A-6G**), the dead space **119a** at the wound site **114** may be calculated based on subtracting the measured volume of the removed fluid canister circuit **202** from the measured volume of the negative pressure circuit **200**, with the volume of the removed fluid canister circuit **202** of the method **800** of FIGS. **8A-8E** being defined by the dead space **103a** of the removed fluid canister **106**, conduit **136** and upstream tubing portion **110a**; and the volume of the negative pressure circuit **200** being defined by the volume of the removed fluid canister circuit **202** (i.e. dead space **103a** of the removed fluid canister **106**, conduit **136** and upstream tubing portion **110a**), the downstream tubing portion **110b**, dead space **119a** of the wound site **114** and the portion of downstream instillation tubing **108b** extending between the drape layer **117** and instillation tubing valve **109**.

(92) According to various embodiments, in embodiments of method **500** in which the determination of the volume of the dead space **119a** at the wound site **114** at step **508** is based on measured parameters related to the removed fluid canister circuit **202** and negative pressure circuit **200** obtained using the method **800** of FIGS. **8A-8E**, step **508** may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** to account for/factor in the known volumes of the downstream tubing portion **110b** and the portion of the downstream instillation tubing **108b** extending between the drape layer **117** and the instillation tubing valve **109** into the determination of the volume of the dead space **119a** at the wound site **114**.

(93) Although, as described above, in some embodiments of method **500**, additional instillation fluid **105** may be delivered at step **516** without first removing any remaining fluid **121** at the wound site **114**, according to other embodiments, it may be desirable to remove fluid **121** from the wound site **114** prior to the delivery of additional instillation fluid **105**.

(94) Referring to FIGS. **9A-9E**, one embodiment of a method **900** of determining an amount of dead space at a wound site **114** which may be used at step **506** of the method **500** of FIG. 5 in embodiments in which it is desired to remove fluid **121** from the wound site **114** prior to instilling additional instillation fluid **105** is illustrated. In particular, according to the method **900** of FIGS. **9A-9E**, any fluid **121** initially at the wound site **114** immediately prior to step **506** (e.g. as shown in FIG. 7) is displaced from the wound site **114** during the method **900** (i.e. step **506**), such the final dead space **119b** into which the additional instillation fluid **121** will be instilled will be greater than the initial dead space **119a** at the wound site that is present immediately prior to the initiation of step **506** by an amount generally corresponding to a volume of the fluid **121** displaced from the wound site **114** to the removed fluid canister **106** during the method **900**.

(95) As shown by the flowchart in FIG. 9A, the method **900** of determining dead space is substantially the same as the method **600** of calculating the dead space **119** upon initial instillation of instillation fluid **105** to the wound site **114** at step **502** (discussed in more detail with reference to FIGS. **6A-6G**). In particular, similar to steps **604** and **606**, the method **900** of FIG. 9A also includes steps **902** and **904** (shown, e.g., in FIGS. 9B and 9C, respectively) during which negative pressure is applied to and removed from the negative pressure circuit **200**. Similar to steps **608** and **610** of the method **600** of FIG. 6A, the method **900** of FIG. 9 also includes steps **906** and **908** (shown, e.g., in FIGS. 9D and 9E, respectively) during which negative pressure is applied to and removed from the removed fluid canister circuit **202**.

(96) However, unlike the method **600** of FIG. **6A** in which the application and subsequent removal of negative pressure to the negative pressure circuit **200** at steps **604** and **608** may be performed either prior to or after the application and subsequent removal of negative pressure to the removed fluid canister circuit **202** of steps **610** and **612**, in the method **900** of FIG. **9A**, the application and subsequent removal of negative pressure to the negative pressure circuit **200** at steps **902** and **904** is performed prior to the application and subsequent removal of negative pressure to the removed fluid canister circuit **202** of steps **906** and **908**. Additionally, whereas, as described above with reference to the method of FIGS. **6A-6E**, according to various embodiments, any range of negative pressures may generally be applied to the negative pressure circuit **200** at step **604** of method **600**, the negative pressure applied to the negative pressure circuit **200** at step **902** of the method **900** of FIG. **9A** must be sufficient to cause the displacement of fluid **121** from the wound site **114** into the removed fluid canister **106**.

(97) Following the completion of step **908**, the controller **118** may be configured to calculate the volume of the final dead space **119b** at the wound site **114** (which corresponds to the maximum volume of additional instillation fluid **105** that may be delivered to wound site **114**) at step **508** of method **500** of FIG. **5**. More specifically at step **508**, after calculating the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** based on the parameters measured at steps **904** and **908** (in a manner similar to that described with reference to step **612** of the method **600** of FIGS. **6A-6G**), the final dead space **119b** at the wound site **114** may be calculated based on subtracting the measured volume of the removed fluid canister circuit **202** from the measured volume of the negative pressure circuit **200**, with the volume of the removed fluid canister circuit **202** of the method **800** of FIGS. **9A-9E** being defined by the final dead space **103b** of the removed fluid canister **106** (with the final dead space **103b** of the removed fluid canister **106** being generally equal to the difference between an initial dead space **103a** within the removed fluid canister **106** and the volume of fluid **121** displaced into the removed fluid canister **106** from the wound site **114** at step **802**, as shown, e.g., in FIG. **9B**), conduit **136** and upstream tubing portion **110a**; and the volume of the negative pressure circuit **200** being defined by the volume of the removed fluid canister circuit **202** (i.e. final dead space **103b** of the removed fluid canister **106**, conduit **136** and upstream tubing portion **110a**), the downstream tubing portion **110b**, final dead space **119b** of the wound site **114** and the portion of downstream instillation tubing **108b** extending between the drape layer **117** and instillation tubing valve **109**.

(98) According to various embodiments, in embodiments of method **500** in which the determination of the volume of the dead space **119** at the wound site **114** at step **508** is based on measured parameters related to the removed fluid canister circuit **202** and negative pressure circuit **200** obtained using the method **900** of FIGS. **9A-9E**, step **508** may also include subtracting or otherwise adjusting the calculated difference between the volumes of the removed fluid canister circuit **202** and the negative pressure circuit **200** to account for/factor in the known volumes of the downstream tubing portion **110b** and the portion of the downstream instillation tubing **108b** extending between the drape layer **117** and the instillation tubing valve **109** into the determination of the volume of the dead space **119a** at the wound site **114**.

(99) In some embodiments of method **500** of FIG. **5** in which fluid **121** from the wound site **114** is removed prior to the instillation of additional instillation fluid **105** at step **516**, it may be desirable to ensure that the initial dead space **103a** in the removed fluid canister **106** immediately prior to beginning the step of determining dead space at the wound site at step **506** is sufficient to hold fluid **121** that will be displaced from the wound site **114** into the removed fluid canister during step **506**, so as to avoid the risk of removed fluid canister **106** overflow.

(100) Accordingly, in some embodiments of method **500** in which fluid **121** from the wound site **114** is removed prior to the instillation of any additional instillation fluid **105** at step **516**, the method of step **506** of determining dead space at the wound site **114** (e.g., such as described with reference to the method **900** of FIGS. **9A-9E**) may include determining whether there is sufficient

dead space at the removed fluid canister **106** to hold the fluid **121** from the wound site **114** that may be displaced into the removed fluid canister **106** as part of the method of determining dead space at the wound site **114**.

(101) Illustrated in FIGS. **10A-10C** is one embodiment of such a method that may be used to minimize the risk of overflow of the removed fluid canister **106** during step **506** in which dead space at the wound site **114** is being determined (e.g., via method **900** as described in FIGS. **9A-9E**). At steps **1002** and **1004** (shown in FIGS. **10B** and **10C**, respectively) negative pressure is applied to and removed from the removed fluid canister circuit **202** to determine the initial dead space **103a** in the removed fluid canister **106** prior to beginning step **506** (e.g. as shown in FIG. **7**). In general, the steps **1002** and **1004** of the method **1000** of FIGS. **10A-10E** may be performed in a manner substantially similar to the manner in which steps **608** and **610** of the method **600** of FIGS. **6A-6G** are performed. At step **1006**, the volume of the removed fluid canister circuit **202** is calculated based on the parameter measured at step **1004** (in a manner similar to that described with reference to step **612** of the method **600** of FIGS. **6A-6G**). Once the volume of the removed fluid canister circuit **202** has been calculated, the known volumes of the conduit **136** and the upstream tubing portion **110a** may be subtracted from the calculated removed fluid canister circuit **202** to determining the volume of the initial dead space **103a** in the removed fluid canister **106** (i.e. the maximum volume of fluid **121** displaced from the wound site **114** that the removed fluid canister **106** may hold).

(102) Once the volume of the initial dead space **103a** has been calculated at step **1006**, at step **1008**, the controller **118** may be configured to estimate the volume of the fluid **121** at the wound site **114** at the time immediately preceding the determination of dead space at the wound site **114** at step **506**. The volume of the fluid **121** at the wound site **114** may be based on any number of different factors and variables such as, e.g., stored values of quantities of instillation fluid **105** previously delivered to the wound site **114**, stored values of fluid **121** previously removed from the wound site, elapsed time (e.g. from a prior instillation, a prior removal of fluid **121**, etc.), etc., with the controller **118** at step **1008** further being configured to compare this estimated volume of fluid **121** to the initial dead space **103a** calculated at step **1006**, alerting the user to empty the removed fluid canister **106** at step **1010** if the controller **118** determines that the estimated fluid **121** volume exceeds the calculated initial dead space **103a**. If the calculated initial dead space **103a** is sufficient to hold the estimated volume fluid **121** from the wound site **114**, at step **1012** the controller **118** may be configured to begin the step **506** of determining dead space at the wound site **114**, e.g., according to method **900** as described with reference to FIGS. **9A-9E**.

(103) As noted above, according to some embodiments of method **500**, it may be advantageous to monitor changes in the volume of the wound site **114** to track the progress of the healing of the wound site **114** at an optional step **510**.

(104) In general, the volume of the wound site **114** is defined by the entirety of the interior extending between the wound site **114** and the drape layer **117** attached to the skin **116** about the wound site **114**. At various points during treatment using the NPWT system **100**, located within and defining the volume of the wound site may be any one of, and any combination of: the wound dressing **112**, fluid **121**, and/or dead space **119**. As will be understood, unless the wound dressing **112** is replaced during treatment, the volume of the wound site **114** volume occupied by the wound dressing **112** will generally remain unchanged over the course of treatment, whereas the portion of the wound site **114** volume occupied by the fluid **121** and/or dead space **119** may change with time.

(105) Referring to FIG. **11**, a block diagram illustrating one embodiment of a method **1100** of tracking wound site **114** healing progression which may be used at step **510** of the method **500** of FIG. **5** is illustrated. At step **1102**, an initial volume of the wound site **114** is estimated and recorded by the controller **118** at a point in time prior to an initial instillation of instillation fluid **105** to the wound site **114**, and may serve as a baseline against which subsequent wound site **114** volume estimates are compared to track healing progress. According to various embodiments, estimation

of the initial volume of the wound site **114** at step **1102** may be performed according to (or as) step **616** of method **600** described with reference to FIGS. **6A-6G**.

(106) At step **1104**, the estimated volume of the wound site **114** is determined and recorded at one or more additional times during treatment (e.g., once per day) following the estimation of the initial wound site **114** volume at step **1102**, with the times at which such one or more wound site **114** volumes are estimated and the values of the determined wound site **114** volume being stored as data points within the memory of therapy device **102** and/or presented to a user as an output of therapy device **102** (e.g., via communications interface **124** or user interface **126**). In some embodiments, the estimated wound volume can be plotted as a function of time.

(107) The additional wound site **114** volume estimates determined at one or more additional times over the course of treatment at step **1104** may be estimated according to any number of different processes. For example, according to some embodiments, the wound site **114** volume estimates recorded at step **1104** may be based on the final dead space volume at the wound site **114** calculated, e.g., at step **508** of method **500** and/or using method **900** as described with reference to FIG. **5** and or **9A-9E**, respectively.

(108) As shown at step **510** of FIG. **5** and step **616** of FIG. **6A**, according to some embodiments, the wound site **114** volume estimates at steps **1102** and/or steps **1104** may be performed in conjunction with method of delivering of instillation fluid **105** to the wound site **114**. However, as will be understood, according to other embodiments the determination of and recording of some, all, or none of the wound site **114** volume estimates at steps **1102** and/or steps **1104** may be performed independent of any delivery of instillation of instillation fluid **105** to the wound site **114**.

(109) As additional wound site **114** volume estimates are obtained at steps **1104**, at step **1106**, changes in the estimated wound site **114** volumes over time may be used to determine healing progression of the wound site **114**. For example, step **1106** may include comparing wound site **114** volume estimates obtained at step **1104** to one or more previous estimates of the wound site **114** volume (obtained at either step **1104** or step **1102**) to identify a change in the wound site **114** volume. In some embodiments, step **1106** may additionally include determining a rate at which the wound site **114** is healing based on the changes in the estimated wound site **114** volume over time. In some embodiments, step **1106** may include extrapolating or predicting a time at which wound site **114** will be fully healed based on the series of wound site **114** volume estimates stored by the controller **118**. For example, step **1106** may include predicting a time at which the estimated wound site **114** volume will reach zero (or another threshold value) based on the initial wound site **114** volume estimate obtained at step **1102** and the series of additional wound site **114** volume estimates obtained at step **1104**.

(110) According to some embodiments, instead of, or in addition to, a calibrated leak system **113** being provided which is located upstream of the tubing valve **111**, the NPWT system **100** may include a calibrated leak system **113** located downstream of the tubing valve **111**. In general, such embodiments in which a calibrated leak system **113** is located downstream of the tubing valve **111** may operate in a manner substantially similar to the various methods described with reference to FIGS. **1-11**. However, in contrast to the step of monitoring the pressure decay within the removed fluid canister circuit **202**, determining a volume of the removed fluid canister circuit **202** based on the monitored pressure decay, and subsequently using the determined volume of the removed fluid canister circuit **202** to calculate wound site **114** volume, in the method **1200** of FIG. **12**, pressure decay is instead monitored within the wound site circuit **204**, with the determined volume of the wound site circuit **204** subsequently being used to calculate the dead space **103** in the removed fluid canister **106**.

(111) For example, referring to FIG. **12**, a flowchart detailing the steps of a method **1200** for an initial set up of NPWT system **100** and delivery of an initial amount of instillation fluid **105** to a wound site **114** is shown according to one embodiment in which a calibrated leak system **113** of the NPWT system **100** is positioned downstream of the tubing valve **111**. In general, steps **1202**, **1204**

and **1206** of the embodiment of the method **1200** of FIG. **12** may be performed in a manner substantially similar to that as described with reference to steps **602**, **604** and **606** of the method **600** of FIG. **6A**.

(112) At step **1208**, the controller **118** is configured to initiate operation of the pump **120** to apply a second negative pressure (which may be equal to or different from the negative pressure applied by the controller **118** at step **1204**) to the negative pressure circuit **200**. According to various embodiments, the instillation tubing valve **109** and the vent valve **113b** may be configured to be set to closed configurations during the application of negative pressure to the negative pressure circuit **200** at step **1208**. In embodiments in which a controller **118** controlled tubeset module **300** is used, the controller **118** may be configured to instruct the tubeset module **300** to effectuate the closing of the instillation tubing valve **109** and/or the vent valve **113b**.

(113) At step **1210**, following the attainment of the desired second negative pressure within the negative pressure circuit **200** (as, e.g., measured and reported to the controller **118** by pressure sensor **115a** and/or pressure sensor **115b**), the tubing valve **111** is closed so as to define a wound site circuit **204**, the operation of the pneumatic pump **120** is stopped, and air from the ambient environment surrounding the therapy device **102** is allowed to flow through the vent **113a** of the and into the wound site circuit **204**. As air from the ambient environment flows into the wound site circuit **204**, parameters related to the flow of air through the vent **113a** and into the wound site circuit **204** are monitored, with the measured parameters subsequently being used by the controller **118** to calculate the volume of the wound site circuit **204** at step **1212**. According to various embodiments, the parameters related to the flow of air into wound site circuit **204** may include, e.g., the rate of flow of air into the wound site circuit **204** (as measured, e.g., by flow detector **113c**), the duration of time required for pressure within the wound site circuit **204** to increase to a predetermined pressure (e.g. ambient pressure) following the opening of the vent **113a** and/or ceasing operation of the pump **120** at step **1210**, the pressure (as, e.g., measured by pressure sensor **115b**) within the wound site circuit **204** as the pressure increases from the negative pressure applied at step **1208** to the predetermined pressure; etc.

(114) At step **1212**, the controller **118** may be configured to determine the volume of the wound site circuit **204** based on the parameters measured during step **1208**. According to some embodiments, the controller **118** may base this wound site circuit **204** volume calculation on stored relationships between various measured parameter values and corresponding volumes. These relationships between measured parameter measurements and corresponding volumes that are stored by the controller **118** may include various functions, models, lookup table, etc., and may be based on pre-existing information input and stored by the controller **118**, or on information obtained and processed by the controller **118** during an optional, initial training procedure conducted by the controller **118** prior to the use of the NPWT system **100** to treat wound site **114** (e.g. prior to the initiation of method **500**; as part of the initial setup and initial instillation of instillation fluid of step **502**; etc.). One non-limiting examples of embodiments of training procedures by which such relationships may be generated by the controller **118** are outlined in related, U.S. Provisional Application 62/650,132, filed Apr. 17, 2018 and titled WOUND THERAPY SYSTEM WITH WOUND VOLUME ESTIMATION, the entire disclosure of which is incorporated by reference herein.

(115) Using the determined volume of the wound site circuit **204**, the controller **118** may determine the volume of the dead space **119** at the wound site **114** (i.e. the portion of the interior space defined between the wound site **114** and the lower surface of the drape layer **117** that is not occupied by the wound dressing **112** and/or any instillation fluid **105**/other fluid) by subtracting or otherwise adjusting the calculated volume of the wound site circuit **204** to account for/factor in the known volumes of the downstream tubing portion **110b** and the portion of the downstream instillation tubing **108b** extending between the drape layer **117** and the instillation tubing valve **109** into the determination of the volume of the dead space **119** at the wound site **114**.

(116) At step **1214**, an initial quantity of instillation fluid **105** that is to be delivered to the wound site **114** is calculated. According to various embodiments, the calculated initial quantity of instillation fluid **105** that is delivered to the wound site **114** may be based on the volume of the dead space **119** calculated by the controller **118** at step **1212**. For example, in some embodiments, the controller **118** may calculate the initial volume of instillation fluid **105** to be delivered to the wound site **114** by multiplying the volume of dead space **119** calculated at step **1212** by a fluid instillation factor. The fluid instillation factor may be equal to or less than one (i.e., between zero and one) such that the volume of instillation fluid **105** delivered to the wound site **114** does not exceed the available space within the drape layer **117** (thereby minimizing inadvertent leakage from the wound dressing **112**/drape layer **117**). In some embodiments, the fluid instillation factor is between approximately 0.2 and approximately 0.8. However, it is contemplated that the fluid instillation factor can have any value in various alternative embodiments.

(117) As noted previously with reference to step **510**, in addition to being used to calculate a quantity of instillation fluid **105** to be delivered during any stage of treatment using NPWT system **100** and under any number of different conditions (e.g. allowing for the calculation of additional instillation fluid **105** to be delivered at step **516** even if the removed fluid canister **106** has been emptied, or entirely replaced with a different sized removed fluid canister **106** during the course of treatment), in some embodiments the NPWT system **100** may be additionally, or alternatively, used to monitor and track the progress of healing of the wound site **114** over time. Accordingly, in some embodiments, at step **1216**, an initial baseline wound site **114** volume estimate may optionally be determined (via, e.g., a method as described with regards to FIG. **11**) and stored by the controller **118**, which may be used as a reference point against which future wound site **114** volume estimates may be compared to track healing progression of the wound site **114**.

(118) At step **1218** the dead space **103** of the removed fluid canister **106** may be calculated to determine whether the dead space within the removed fluid canister **106** will be sufficient to collect any fluids **121** from the wound site **114** (including non-absorbed instillation fluid **105**) following the delivery of instillation fluid **105** at step **516**. As will be understood, in embodiments in which the NPWT system **100** has not been operated prior to the use of the NPWT system **100** at step **1202**, the volume of the removed fluid canister **106** should be empty, such that the dead space **103** of the removed fluid container **106** should be equal to the volume of the removed fluid canister **106**.

(119) The dead space **103** of the removed fluid container may be calculated by subtracting the known volumes of conduit **136** and the upstream tubing portion **110a** from a volume of the removed fluid canister circuit **202** determined by subtracting the volume of the wound site circuit **204** calculated at step **1212** from a determined volume of the negative pressure circuit **200**. As will be understood, the volume of the negative pressure circuit **200** may be determined in a manner similar to the method via which the volume of the wound site circuit **204** is determined at step **1212**.

(120) Similar to step **514**, at step **1220** an alarm may be presented to a user if the initial volume of instillation fluid **105** to be delivered calculated at step **1214** exceeds the dead space **103** of the removed fluid canister **106**. Otherwise, if the volume of the initial instillation fluid **105** to be delivered does not exceed the dead space **103** of the removed fluid canister **106**, the calculated instillation fluid **105** is delivered to the wound site **114** at step **1222**.

(121) As will be understood, in some NPWT system **100** embodiments in which a calibrated leak system **113** is provided both upstream and downstream of the tubing valve **111**, the NPWT system **100** may be operated to estimate wound site **114** volume and/or estimate dead space **103** at the removed fluid canister **106** according to a method that is the same as or similar to the method **600** of FIG. **6A** or a method that is the same as or similar to the method **1200** of FIG. **12**.

(122) In other embodiments of NPWT system **100** in which both an upstream and downstream calibrated leak system **113** are provided, the NPWT system **100** may be operated to estimate wound

site **114** volume and/or estimate dead space **103** at the removed fluid canister **106** according to a method that is the same as or similar to the method **600** of FIG. **6A** and a method that is the same as or similar to the method **1200** of FIG. **12**. For example, according to some embodiments, a modified method of operating a NPWT system **100** having both upstream and downstream calibrated leak systems **113** may include the steps of: monitoring pressure decay within the negative pressure circuit **200** (such as, e.g., described with reference to step **606** of the method **600** of FIG. **6** and/or step **1206** of the method **1200** of FIG. **12**); monitoring pressure decay within the removed fluid canister circuit **202** (such as, e.g., described with reference to step **610** of the method **600** of FIG. **6**); and monitoring pressure decay within the wound site circuit **204** (such as, e.g., described with reference to step **1210** of the method **1200** of FIG. **12**).

(123) In such embodiments, the determination of wound site **114** volume based on direct measurement (e.g. using the method **1200** of FIG. **12**) may be compared to wound site **114** volume calculated based on indirect measurement (e.g. using the method **600** of FIG. **6A**) and the determination of dead space **103** at the removed fluid canister **106** based on direct measurement (e.g. using the method **600** of FIG. **6A**) may be compared to dead space **103** volume calculated based on indirect measurement (e.g. using the method **1200** of FIG. **12**). The controller **118** in such embodiments may be configured to generate an alarm or alert in response to a discrepancy between the direct and indirect measurements of wound site **114** volume and/or dead space **103** at the removed fluid canister **106**. By providing such redundancy to the wound site **114** and/or removed fluid canister **106** dead space **103** calculations, such embodiments may be configured to allow the NPWT system **100** to provide more accurate and reliable results.

(124) As will be understood, according to various embodiments, the controller **118** may be programmed to allow the NPWT system **100** to determine volume relative to the wound site **114** using any or all of the methods described herein. Accordingly, while in some embodiments the controller **118** may optionally be preprogrammed to automatically determine a volume of instillation fluid **105** to be delivered according to a particular method (e.g. the method **900** embodiment illustrated in FIGS. **9A-9E**), the controller **118** may optionally also allow a user to select any of the other modes of calculating a volume relative to the wound site **114** based on whether the user desires to, e.g.: remove fluid **121** from the wound site **114** prior to instillation of additional instillation fluid **105**; verify sufficient dead space **103a** in the removed fluid canister **106** prior to determining the dead space at the wound site **114**; verify sufficient dead space **103b** in the removed fluid canister **106** prior to the instillation of a calculated quantity of additional instillation fluid **105** to be delivered to the wound site **114**; monitor changes in the wound site **114** volume to track healing progression; etc.

Tubeset Module

(125) Although in some arrangements, some or all of the calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109**, or other NPWT system **100** components may be configured to be manually operated/actuated/utilized by a user, as noted above, according to various embodiments, some or all of these components may alternatively be configured to be operated/actuated/utilized by the controller **118**, without requiring any user assistance to do so. In such a manner, implementation of the system for/method of determining a volume of instillation fluid to be delivered to a wound site, estimating a volume of a wound, monitoring healing progression of a wound, and/or other use of the NPWT system **100** may be fully automated using the controller **118**, allowing for easier use of the NPWT system **100**.

(126) By providing the NPWT system **100** with an automated manner via which controller **118** may control or otherwise interact with one or more of the calibrated leak system **113**, tubing valve **111**, instillation tubing valve **109**, and/or other component(s) of the NPWT system **100**, the tubeset module **300** may increase the accuracy of the NPWT system **100**. For example, in light of the ability of the controller **118** to utilize the tubeset module **300** to independently actuate (i.e. without user intervention) the operation of the calibrated leak system **113**, the tubing valve **111** and/or the

instillation tubing valve **109** elements, the controller **118** may be configured to increase the rate at which data related to instillation fluid volume estimation, wound site volume estimation, wound site **114** healing progression monitoring, and/or other functions of the NPWT system **100** is gathered. By increasing the data points used to provide such information, the reliability of the information provided by the controller **118** may thereby be increased. Similarly, the obviation or minimization of user involvement provided by the tubeset module **300** may facilitate (and thereby increase the likelihood of) the usage of a dual calibrated leak system **113** arrangement as described with reference to FIG. **12** above, thus also increasing the reliability of the NPWT system **100**.

(127) In general, the tubeset module **300** comprises a housing element **304** containing a power source **301**, a communications interface **302**, and one or more actuatable elements **303** configured to be controlled by the controller **118**. In some embodiments, the tubeset module **300** may optionally also comprise one or more additional non-actuatable elements **305**, such as, e.g., pressure sensor **115a** and/or pressure sensor **115b**. According to embodiments in which calibrated leak system **113** is not defined by a vent valve **113b** and only comprises a non-actuatable vent **113a**, the non-actuatable element(s) **305** may comprise such a calibrated leak system **113** formed without a vent valve **113b**.

(128) As will be understood, according to some embodiments, some or all of the actuatable elements **303** may be configured so as to be self-actuating. In some such embodiments, the actuatable element **303** may comprise an internal actuator that is operably connected (via a wired, wireless, or any other type of connection) to the power source **301** and/or communications interface **302** of the tubeset module **300**, via which instructions received from the controller **118** and/or power are relayed to the actuator of the actuatable element **303**. In other such embodiments, such self-actuation actuatable element **303** may individually comprise one or both of a power source and/or communications interface (in addition to the power source **301** and/or communications interface **302** of the tubeset module). In such embodiments, the instructions from the controller **118** may be received directly by the communication interface of the actuatable element **303** from the controller, or may be received indirectly by the communication interface of the actuatable element **303** from the communication interface **302** of the tubeset module **300**. In other embodiments, some or all of the actuatable elements **303** may be configured to be actuated by any number of different types of, or combinations of known actuators that are contained by the housing element **303**, with the actuators of the housing element **303** being configured to effectuate actuation of the one or more actuatable elements **303** in response to instructions received from the controller **118**.

(129) The power source **301** may comprise any number of, and combination of, sources of energy that are configured to supply sufficient energy to the communications interface **302**, actuatable element(s) **303** and/or non-actuatable elements **305** contained by the housing element **304** as required for operation of the NPWT system **100**. In some embodiments in which some or all of the tubeset module **300** is integrated into the therapy device **102**, the power provided by the power source **301** of the housing element **304** may comprise a power source of the therapy device **102**.

(130) The communications interface **302** may comprise any number of, and combination of, wired and/or wireless connections via which the tubeset module **300** may receive communications (such as, e.g., actuation signals) from the controller **118**. According to some embodiments, the communications interface **302** may optionally also be configured to send information to and/or receive information from the controller **118**, other tubeset module **300** housing elements **304** (such as, e.g., information regarding the status of the one or more actuatable elements **303** and/or non-actuatable elements **305** of the tubeset module **300**), and/or other sources. In some embodiments in which some or all of the tubeset module **300** is integrated into the therapy device **102**, the communications interface **302** of the housing element **304** may be defined by a portion of a communications interface of the therapy device **102**.

(131) According to some arrangements, tubeset module **300** may be defined by a single housing

element **304**, with each of actuable elements **303** (e.g. upstream and/or downstream calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109**, etc.) and/or non-actuable elements **305** that are to be controlled/utilized by the controller **118** forming a part of the single, integral housing element **304**. In other embodiments, the tubeset module **300** may be defined by a plurality of separate and distinct housing elements **304**, with each housing element **304** formed with one or more of the various actuable elements **303** and/or non-actuable element **305** that are to be controlled/utilized by the controller **118**.

(132) According to various arrangements, the one or more housing elements **304** defining the tubeset module **300** may be provided as a separate, discrete, individual component of the NPWT system **100**, which may subsequently be attached to or otherwise incorporated into one or more of the other components of a new or existing NPWT system **100**. In other arrangements, some or all of the one or more housing elements **304** defining the tubeset module **300** may be provided as an integrated part of one or more of the other components of the NPWT system **100**.

(133) For example, in some arrangements, some or the entirety of the tubeset module **300** may be integrated into the wound dressing **112**, with the portion of the tubeset module **300** provided with the wound dressing **112** being configured to be removed from the NPWT system **100** with the removal of the wound dressing **112**. Upon removal of the integrated wound dressing **112**/tubeset module **300**, the entire wound dressing **112**/tubeset module **300** may be disposed of. Alternatively, the tubeset module **300** may be removed from the wound dressing **112** prior to disposal of the wound dressing **112** and optionally reused with another wound dressing **112** and/or other NPWT system **100** component.

(134) In other arrangements, some or the entirety of the tubeset module **300** may be integrated into the removed fluid canister **106**, with the portion of the tubeset module **300** provided with the removed fluid canister **106** being removed from the NPWT system **100** with the removal of the of the removed fluid canister **106** from the NPWT system **100**. In some such embodiments, the tubeset module **300** may be monolithically formed with the removed fluid canister **106**, while in other embodiments; the tubeset module **300** may be non-integrally formed with the removed fluid canister **106**.

(135) According to another arrangement, the tubeset module **300** may be configured to be integrated in-line with one or both of the tubing **108** and/or **110**. In such embodiments, attachment adapters **400** may be provided on one or both of the tubeset module **300** and/or tubing **108** and/or **110** to facilitate a fluid tight attachment of the tubeset module **300** to the tubing **108** and/or **110**. According to some embodiments, the attachment adapters **400** may be provided on the tubeset module **300**, with the attachment adapters **400** being configured to be able to form a fluid tight attachment directly with one or both of the tubing **108** and/or **110**, allowing NPWT systems formed without a tubeset module **300** and/or calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109** to be retrofitted with a tubeset module **300** so as to provide a NPWT system **100** as disclosed herein.

(136) In some arrangements, some or the entirety of the tubeset module **300** may be integrated into the housing of the therapy device **102**. In such embodiments, the efficiency of using the NPWT system **100** may be increased, as by incorporating a tubeset module **300** including some or all of the calibrated leak system **113**, tubing valve **111** and/or the instillation tubing valve **109** into the housing of the therapy device **102**, the time to setup the NPWT system **100** may be reduced as compared to the time that would otherwise be required to setup up a NPWT system **100** in which some or all of the calibrated leak system **113**, tubing valve **111** and/or instillation tubing valve **109** were provided as separate and discrete components of the NPWT system **100**. Additionally, by incorporating the tubeset module **300** into the housing of the therapy device **102**, a NPWT system **100** as described herein may be provided irrespective of the particular tubing, removed fluid canister, wound dressing, or other component(s) that are provided to define a NPWT system **100** for treatment of a wound site **114**.

(137) Referring to FIGS. **13-16**, various embodiments of a tubeset module **300** configured to allow for partially or fully automated control of the NPWT system **100** using the controller **118** are shown. As will be understood, although reference has been made to the controller **118** being provided as part of the therapy device **102**, it is to be understood that, according to various arrangements, the controller **118** may be provided separate and remote from the therapy device **102** and/or NPWT system **100** (e.g., by a remote medical provider). In such embodiments, the remotely provided controller **118** may be configured to communicate directly with the tubeset module **300** and/or indirectly with the tubeset module **300** via a communications interface provided by the therapy device **102**.

(138) As illustrated by the NPWT system **100** embodiment of FIG. **13**, in some arrangements, the tubeset module **300** is provided as a single, integrated housing element **304** containing actuatable elements **303** comprising a calibrated leak system **113**, a tubing valve **111** and an optional instillation tubing valve **109**. According to some embodiments, one or both of the tubing valve **111** and the optionally provided instillation tubing valve **109** may comprise the same or distinct clamps. As shown in FIG. **13**, also contained within the housing element **304** is a power source **301** configured to actuate the actuatable element **303** in response to instructions being received from the controller **118** via the communications interface **302**.

(139) Although in the embodiment illustrated in FIG. **13** a single, integral tubeset module **300** is shown as being in-line with both tubing **108** and tubing **110**, according to other arrangements (now shown) it is to be understood that a first housing element **304** comprising the calibrated leak system **113** and a tubing valve **111** may be provided in-line with tubing **110**, while an optional, second housing element **304** comprising instillation tubing valve **109** may be provided in-line with tubing **110**.

(140) As illustrated by the NPWT system **100** of FIG. **13**, in some embodiments, the tubeset module **300** may be formed integrally with upstream tubing portion **110a** and/or upstream instillation tubing **108a**. According to some such embodiments, the upstream tubing portion **110a** and/or upstream instillation tubing **108a** with which the tubeset module **300** is integrally formed may in turn be formed integrally with the therapy device **102**. In such embodiments, the tubeset module **300** is configured to be removably attached to the downstream tubing portion **110b** and/or downstream instillation tubing **108b** formed integral with the wound dressing **112**, such that following use of the NPWT system **100** with a first wound dressing **112**, the therapy device **102** with integrated upstream tubing portion **110a** and/or upstream instillation tubing **108a** and tubeset module **300** may be reused with a new, second wound dressing **112**. In other embodiments, such as, e.g., illustrated by the NPWT system **100** of FIG. **14**, some or all of the tubeset module **300** may alternatively be formed integrally with the wound dressing **112**, with the tubeset module **300** being configured to be removed from the NPWT system **100** with the removal of the wound dressing **112**.

(141) Referring to the NPWT system **100** of FIG. **15**, according to some embodiments, the tubeset module **300** may comprise a first housing element **304** containing a calibrated leak system **113** and an optional instillation tubing valve **109**, pressure sensor **115a** and/or pressure sensor **119** positioned in-line with the tubing **108** and/or **110**. A second housing element **304** comprising a tubing valve **111** may be spaced from the first housing element **304**. As shown in FIG. **14**, according to some arrangements, the second housing element **304** may be integrated into the removed fluid canister **106**. In other embodiments, the second housing element **304** may alternatively be incorporated into the therapy device **102**, or may be provided at a second location in-line with the tubing **110**.

(142) Referring to FIG. **16A**, a block diagram of a NPWT system **100** according one embodiment is shown. As illustrated by the NPWT system **100** of FIG. **16A**, according to some embodiments, fluid communication between some or all of the negative pressure circuit **200** and the ambient environment may be provided by a purge valve system **450** provided along the instillation tubing **108** as an alternative to, or in addition to, calibrated leak **113**. According to various embodiments,

the purge valve system **450** may comprise a structure similar to that of calibrated leak system **113** (including calibrated leak system **113** embodiments comprising any combination of vent **113a**, vent valve **113b** and/or flow detector **113c** components).

(143) As also illustrated by FIG. **16A**, in such NPWT system **100** embodiments, the tubing valve **111** and/or instillation tubing valve **109** may be replaced by a valve assembly **460** that is fluidly attached to the tubing **110** at a junction between the upstream tubing portion **110a** and downstream tubing portion **110b** and that is attached to the instillation tubing assembly at a junction between the upstream tubing **108a** and downstream tubing **108b**, and which is actuatable to a variety of positions. In a first position, the valve assembly **460** may permit fluid flow from the pneumatic pump **120** to the wound site **114** via tubing **110** and from the instillation pump **104** to the wound site **114** via instillation tubing **108**. In a second position, the valve assembly **460** may permit fluid flow from the pneumatic pump **120** to the wound site **114** via tubing **110** while blocking fluid flow from the instillation pump **104** to the wound site **114** via instillation tubing **108**. In a third position, the valve assembly **460** may block fluid flow from the pneumatic pump **120** to the wound site **114** via tubing **110** while permitting flow from the instillation pump **104** to the wound site **114** via instillation tubing **108**. In a fourth position, the valve assembly **460** may fluidly connect the upstream tubing portion **110a** with the upstream instillation tubing **108a**, resulting in the isolation of the downstream tubing portion **110b**, the downstream tubing **108**, and the wound dressing **112** from the remainder of the therapy device **102**.

(144) When in the first configuration, the valve assembly **460** defines a negative pressure circuit **200** is defined by the tubing **136**, the fluid canister **106**, the tubing **110**, the wound site **114** and the portion of the instillation tubing extending between the wound site **114** and the purge valve **450**. When in the fourth configuration, the valve assembly **460** defines a removed fluid canister circuit **202** is defined by the tubing **136**, the fluid canister **106**, the upstream tubing portion **110a**, and the portion of the upstream tubing **108a** extending between the valve assembly **460** and the purge valve **450** and a wound site circuit **204** defined by the downstream tubing portion **110b**, the wound site **114**, and the downstream tubing **108b**.

(145) As will be understood, the valve assembly **460** and the purge valve **450** of the NPWT system **100** of FIG. **16A** may be operated in a manner similar to the operation of the tubing valve **111**, calibrated leak, and/or instillation tubing valve **109** as described with reference to any of the methods described herein for determining wound site volume, estimating a volume of fluid to be instilled, monitoring wound healing progression, or performing any other functions using the NPWT system **100**.

(146) Referring to FIG. **16B**, a tubeset module **300** configured for used with a NPWT system **100** incorporating a purge valve **450** (such as, e.g. shown in FIG. **16A**) is shown according to one embodiment. As illustrated by FIG. **16B** embodiments in which the purge valve **450** is provided as a discrete component of the therapy device **112** capable of being automatically actuated by the controller **118**, the tubeset module **300** may comprise only a single actuatable element **303**, defined by valve assembly **460**. As will be understood, in other embodiments, (such as, e.g. where the purge valve **450** provided as part of the therapy device is not automatically actuatable by the controller **118**), the purge valve **450** may be provided as a part of a tubeset module **300** that is partially or entirely integrated into the therapy device **112**.

(147) Although in the NPWT system **100** embodiment illustrated in FIG. **16A** the purge valve **450** is illustrated as being provided as part of the therapy device **112**, according to other embodiments, the purge valve **450** may be alternatively, or additionally, provided as a part of the upstream tubing **108a**. In such embodiments, the purge valve **450** may accordingly be provided as an actuatable element **303** of the tubeset module **300**.

(148) As will be understood, the controller **118** may be configured to effectuate any number of different operations using the NPWT system **100** based on the selective, fully automated actuation of/interaction with some or all of the actuatable elements **303** and/or non-actuatable elements **305**

of a tubeset module **300** according to any number of different methods and protocols. According to various embodiments, the order and/or combination of instructions transmitted by controller **118** to the tubeset module **300** and/or the information received by the controller **118** from the tubeset module **300** may be configured to automatically operate the tubeset module **300** in a manner that allows the controller **118** to automatically effectuate one or more of the methods **500**, **600**, **800**, **900**, **1000**, **1100**, **1200**, etc. described herein.

(149) Represented in FIG. **17** is one method **1700** via which the controller **118** may utilize a tubeset module **300** containing actuable elements **303** comprising a tubing valve **111**, instillation tubing valve **109**, and calibrated leak system **113** and non-actuable element(s) **305** comprising one or both of pressure sensor **115a** and/or pressure sensor **115b** to automatically control the NPWT system **100** to determine dead space **119** at a wound site **114** according to a method such as, e.g., described with reference to the method **500** of FIG. **5** and the method **600** of FIG. **6A**.

(150) At step **1701**, in response to the controller **118** being initiated to determine dead space at a wound site **114** (such as, e.g., at step **506** of the method **500** of FIG. **5**), the controller **118** may initiate communication with the tubeset module **300** to confirm that the instillation tubing valve **109** and vent valve **113b** of the calibrated leak system **113** are closed, and that the tubing valve **111** is opened. If the instillation tubing valve **109** and/or vent valve **113b** are open, the controller **118** may instruct the tubeset module **300** to effectuate actuation of the instillation tubing valve **109** and/or vent valve **113b** into a closed configuration. Similarly, if the tubing valve **111** is detected by the controller **118** as being closed, the controller **118** may transmit instructions to the tubing valve **111** via the communications interface **302** to actuate opening of the tubing valve **111**.

(151) Once the controller **118** has received, via the communications interface **302**, confirmation that the instillation tubing valve **109** and vent valve **113b** are closed and the tubing valve **111** is open, the controller **118** may be configured to initiate operation of the pneumatic pump **120** to apply negative pressure to the negative pressure circuit **200** (such as, e.g., described with reference to step **604** of the method **600** of FIG. **6A**). During operation of the pneumatic pump **120**, the controller **118** at step **1703** may be configured to receive from the pressure sensor **115a** and/or pressure sensor **115b** pressure readings corresponding to the pressure within the negative pressure circuit **200**. As will be understood, the pressure readings received by the controller **118** at step **1703** may be received continuously, at predetermined intervals, and/or in response to specific requests for pressure readings transmitted by the controller **118** to the tubeset module **300** via communications interface **302**.

(152) In response to receiving pressure readings from the tubeset module **300** indicative of the pressure within the negative pressure circuit **200** having reached a threshold pressure, the controller **118** at step **1705** may be configured to stop operation of the pneumatic pump **120** and transmit to the tubeset module **300** an actuation signal configured to cause the opening of the vent valve **113b**.

(153) At step **1707**, the controller **118** may be configured to receive from the pressure sensor **115a** and/or pressure sensor **115b** pressure readings corresponding to pressure decay within the negative pressure circuit **200**, such as, e.g., described with reference to step **606** of FIG. **6A**. The pressure readings received by the controller **118** at step **1707** may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller **118** to the tubeset module **300** via communications interface **302**.

(154) Once the controller **118** has received pressure readings from the tubeset module **300** indicative of the pressure within the negative pressure circuit **200** having reach a threshold pressure (such as, e.g., ambient pressure), the controller **118** at step **1709** may be configured to effectuate, using the tubeset module **300**, the actuation of the closing of the tubing valve **111** and the vent valve **113b** in advance of the application of negative pressure to the resultant removed fluid canister circuit **202** (such as, e.g., during step **608** of the method **600** of FIG. **6A**).

(155) At step **1711**, the controller **118** once again may be configured to receive pressure readings from the tubeset module **300**. The pressure readings received by the controller **118** at step **1711**

may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller **118** to the tubeset module **300** via communications interface **302**. In response to receiving pressure readings from the tubeset module **300** indicative of the pressure within the removed fluid canister circuit **202** having reached a threshold pressure, the controller **118** at step **1713** may be configured to stop operation of the pneumatic pump **120** and transmit to the tubeset module **300** an actuation signal configured to cause the opening of the vent valve **113b**.

(156) At step **1715**, the controller **118** may be configured to receive from the pressure sensor **115a** pressure readings corresponding to pressure decay within the removed fluid canister circuit **202**, such as, e.g., described with reference to step **610** of FIG. **6A**. The pressure readings received by the controller **118** at step **1715** may be received continuously, at predetermined intervals, or in response to specific requests for pressure readings transmitted by the controller **118** to the tubeset module **300** via communications interface **302**.

(157) According to some embodiments, following step **1715**, at step **1717**, the controller **118** may be configured to actuate, using the tubeset module **300**, the opening of the instillation tubing valve **109**, in advance of the instillation of instillation fluid to the wound site **114** (such as, e.g., described with reference to step **516** of the method **500** of FIG. **5** and/or step **622** of the method **600** of FIG. **6A**).

Wound Therapy System with Internal Alternating Orifice

(158) Referring now to FIG. **18**, an NPIWT system **2100** is shown, according to an exemplary embodiment. The NPIWT system **2100** includes a dressing **2102** fluidly communicable with a canister **2104** via first tubing **2106** and a therapy unit **2108** coupled to the canister **2104**. As shown in FIG. **18**, the NPIWT system **2100** also includes an instillation fluid source **2110** fluidly communicable with the dressing **2102** via the therapy unit **2108** and second tubing **2112**. The NPIWT system **2100** and components thereof may correspond to, may be implemented with, may be combined, and/or may otherwise provide various features of the systems and methods described above with reference to FIGS. **1-17**. It should be understood that the present disclosure contemplates various combinations of the embodiments shown in the drawings.

(159) The dressing **2102** is shown as applied to a wound bed **2114**. The dressing **2102** includes a drape **2116** sealed over the wound bed **2114** and a foam layer **2118** positioned between the drape **2116** and the wound bed **2114**. In various embodiments, the dressing **2102** may include various layers and features. The drape **2116** may be made of a substantially air-impermeable material (e.g., a polyurethane-based material) and may include an adhesive border that allows the drape to be sealed to a patient's skin around the wound bed **2114**. The foam layer **2118** may include a manifold layer that allows airflow therethrough and facilitates the distribution of negative pressure across the wound bed **2114**. A wound space **2120** that includes the open volume (i.e., through which air may flow) in the foam layer **2118** and otherwise situated between the drape **2116** and the wound bed **2114** is thereby established.

(160) The first tubing **2106** extends from the dressing **2102** to the canister **2104**. A cross-section of the first tubing **2106** is shown in FIG. **21**, according to an exemplary embodiment. As described in detail with reference to FIG. **21**, the first tubing **2106** includes an inner lumen **2400** and one or more outer lumens **2402**. The inner lumen **2400** provides for the flow of fluid from the wound space **2120** into the canister **2104**. The one or more outer lumens **2402** are fluidly communicable with a pressure sensor **2124** to facilitate measurement of the pressure at the wound space **2120**. The one or more outer lumens **2402** are also fluidly communicable with a valve **2126** as described below. It should be understood that, while described as inner and outer in the examples herein, any geometrical arrangement of multiple lumens may be used in various embodiments. A connection pad (e.g., low pressure interface) **2121** is coupled to the drape **116** and facilitates connection of the first tubing **106** to the dressing **2102**.

(161) The canister **2104** is configured to collect wound exudate (e.g., fluid, other debris) removed

from the wound space **2120** via the first tubing **2106**. The canister **2104** is fluidly communicable with the wound space **2120** via the first tubing **2106**. The canister **2104**, the first tubing **2106**, and the dressing **2102** thereby define a sealed space that includes the wound space **120**.

(162) The therapy unit **2108** is coupled to the canister **2104** and includes a pneumatic pump **2122** fluidly communicable with the sealed space, a sensor **2124** configured and positioned to measure pressure in the sealed space, a valve **2126** positioned between the sealed space and an environment, a user interface **2128**, and an instillation pump **2130** coupled to the second tubing **2112**. The therapy unit **2108** also includes a control circuit **2132** communicably and operably coupled (e.g., capable of exchanging electronic signals with) the pneumatic pump **2122**, the sensor **2124**, the valve **2126**, the user interface **2128**, and the instillation pump **2130**.

(163) The pneumatic pump **2122** is controllable by the control circuit **2132** and operable to pump (e.g., draw, remove) air from the canister **2104**, the first tubing **2106**, and the wound space **2120** (i.e., from the sealed space). The pneumatic pump **2122** may thereby create a negative pressure in the sealed space relative to atmospheric pressure, for example between 25 mmHg and 175 mmHg. The pneumatic pump **2122** may create a pressure differential that causes fluid and debris to be drawn out of the wound space **2120**, through the first tubing **2106**, and into the canister **2104**.

(164) The sensor **2124** is positioned and configured to measure the pressure in the sealed space. As shown in FIG. **18**, the pressure sensor **2124** is positioned to measure pressure via one or more outer lumens **2402**. In other embodiments a sensor **2124** may be included to measure pressure elsewhere in the sealed space (e.g., in the canister **104**). The sensor **2124** provides pressure measurements to the control circuit **2132** (e.g., digital values, analog signals). The control circuit **2132** may be configured to receive the pressure measurements from the sensor **2124** and use the pressure measurements in a control loop to generate control signals for the pneumatic pump **2122** that cause the pneumatic pump **2122** to maintain a desired pressure in the sealed space or provide a desired pattern of pressure in the sealed space.

(165) The user interface **2128** may include a display screen, a touch screen, a speaker, a button, a switch, or any other element capable of providing information to a user or receiving input from a user. In some embodiments, the control circuit **2132** is configured to generate a graphical user interface and cause the graphical user interface to be displayed on the user interface **2128**. The graphical user interface may include various information about the NPIWT provided by the NPIWT system **2100**, for example relating to the pressure in the sealed space, an amount of instillation fluid to be provided, a schedule of negative pressure and instillation cycles, and/or a size of the wound space **2120**. The user interface **2128** may allow a user to input commands and settings relating to the operation of the therapy unit **2108**. The control circuit **2132** may receive such inputs from the user interface **2128** and control the therapy unit **2108** in accordance with the inputs.

(166) The instillation pump **2130** is configured to cause instillation fluid to be transported from the instillation fluid source **2110** to the wound space **2120** via second tubing **2112**. The instillation pump **2130** may be controllable by the control circuit **2132** to provide a desired amount of the instillation fluid to the wound space **2120**, provide instillation fluid to the wound space **2120** at a desired rate, prevent instillation fluid from flowing to the wound space **2120**, or otherwise control the flow of instillation fluid to the wound space **2120**. The instillation pump may include a peristaltic pump or some other type of pump.

(167) The valve **2126** is controllable between an open position and a closed position. As shown in FIG. **18**, the valve **2126** is located at an interior of the therapy unit **2108** in pneumatic communication with a surrounding environment (e.g., ambient air) via a vent **2134** positioned along an exterior of the therapy unit **2108**. The valve **2126** is also shown as communicable with the one or more outer lumens **2402** of the first tubing **2106**. A filter **2138** is located between the canister **2104** and the pump **2122**. When the valve **2126** is in the open position, air may flow between the surrounding environment and the sealed space via the filter **2138**. When the valve

2126 is in the closed position, air is prevented from flowing therethrough. As shown in FIGS. **19-20** and described in detail with reference thereto, the valve **2126** may be a solenoid valve. In various other embodiments, other types of valves may be included. As described in detail below, the valve **2126** may be controllable to allow a sudden surge (“blast”) of air therethrough in a manner intended to clear a blockage in the one or more outer lumens **2402** of the first tubing **2106**. The valve **2126** may also be controllable to allow a controlled rate of airflow therethrough to facilitate determination of a volume of the wound space **2120**.

(168) The filter **2138** is configured to prevent contaminants from moving from the surrounding environment to the wound space **2120** via the valve **2126** and the one or more outer lumens **2402**. The filter **2138** thereby protects the wound **2114** from infection or other complications. The filter **2138** restricts the rate of flow of air from the surrounding environment into the sealed space through the filter **2138** (e.g., by creating a pressure drop across the filter **2138** due to the filter media, contaminants trapped in the filter media, etc.) to a maximum of a restriction rate of the filter **2138**. The restriction rate may be difficult to ascertain, may vary over time, or may be different in different instances of the filter **2138** (i.e., differing across multiple therapy units **108**).

(169) In the embodiments shown, the restriction rate of the filter is less than a typical rate of airflow through the valve **2126** when the valve **2126** is held in the open position for an extended amount of time (e.g., 500 milliseconds or greater). Accordingly, the difficulty in determining the restriction rate of the filter **2138** leads to a difficulty in determining a rate of airflow into the sealed space when the valve **2126** is held in the open position for an extended amount of time.

(170) The control circuit **2132** is configured to control the operation of the therapy device **2108**. For example, as described in detail below, the control circuit **2132** is configured to control the pneumatic pump **2112** to remove air from the sealed space to establish a negative pressure in the sealed space, control the valve **2126** to provide a controlled leak to the sealed space, receive pressure measurements from the sensor **2122**, determine a volume of the wound space **2120** based on the pressure measurements, and customized a wound therapy based on the volume of the wound space **2120**. In some embodiments, the control circuit **2132** is also configured to detect a potential blockage of a lumen of the first tubing **2106**, control the valve **2126** to the open position to allow a blast of air therethrough, keep the valve **2126** open while the blast of air clears the blockage, and control the valve **2126** to return to the closed position. These and other features of the control circuit **2132** are described in detail below.

(171) Referring now to FIGS. **19-20**, cross-sectional views of the valve **2126** are shown, according to exemplary embodiments. In the embodiments shown, the valve **2126** is a solenoid valve. FIG. **19** shows the valve **2126** in the closed position and FIG. **20** shows the valve in the open position. It should be understood that FIGS. **19-20** show one of many possible embodiments of the valve **2126**.

(172) The valve **2126** includes an inlet **2200** pneumatically communicable with the surrounding environment via the vent **2134**, an outlet **2202** pneumatically communicable with the sealed space via channel **2136**, a solenoid **2206**, a plunger **2204** extending axially through the solenoid **2206** and substantially centered in the solenoid, a stopper **2205** coupled to the plunger **2204**, and a spring **2208** coupled to the plunger **2204**. The solenoid **2206** has a positive lead **2210** and a negative lead **2212** shown as operably coupled (e.g., conductively coupled) to the control circuit **2132**.

(173) The solenoid **2206** includes a coil of wire through which the plunger **2204** extends. When a current flows through the solenoid (e.g., when a voltage differential is applied across the solenoid **2206**, a magnetic field is created in the solenoid **2206**. The magnetic field is substantially aligned with a central axis of the solenoid. The plunger **2204** is made of a magnetic material, such that the magnetic field causes movement of the plunger **2204** when voltage is applied across the solenoid **2206**.

(174) As shown in FIG. **19**, a voltage of approximately zero volts is applied across the solenoid **2206**. That is, the control circuit **2132** prevents a voltage difference between the positive lead **2210** and the negative lead **2212**. Accordingly, approximately zero current is created in the solenoid

2206, and approximately zero magnetic field is created by the solenoid **2206**. The spring **2208** exerts a force on the plunger **2204** that holds the stopper **2205** adjacent the inlet **2200**. The stopper **2205** prevents air from entering the valve **2206** via the inlet **2200**. Airflow from the vent **2134** to the channel **2136** is thereby prevented (i.e., the valve **126** is in the closed position).

(175) As shown in FIG. **20**, a non-zero voltage is applied across the solenoid (e.g., approximately 5 volts). That is, the control circuit **2132** provides a control signal to the valve **2126** by creating a voltage differential between the positive lead **2210** and the negative lead **2212** of the solenoid **2206**. It should be understood that, in various embodiments, various values of the non-zero voltage may be required to operate the valve **2126**. When the control circuit **2132** provides a non-zero voltage to the valve **2126** (i.e., across the solenoid **2206**), a magnetic field is created that causes the plunger **2204** to compress the spring **2208** and move the stopper **2205** away from the inlet **2200**. Air may then flow from the vent **2134** through the valve **2126** to the sealed space (i.e., the valve **2126** is in the open position).

(176) When the non-zero voltage is removed (i.e., when the voltage differential between the positive lead **2210** and the negative lead **2212** is brought to approximately zero), the magnetic field goes to zero and the spring **2208** forces the plunger **2204** and stopper **2205** back to the closed position shown in FIG. **19**. Thus, the valve **2126** may be controlled to repeatedly alternate between the closed position shown in FIG. **19** and the open position shown in FIG. **20** by alternating between an approximately zero voltage and an approximately non-zero voltage. Example voltage patterns for controlling the valve **2126** to provide a controlled rate of airflow therethrough are described in detail below.

(177) Referring now to FIG. **21**, a cross-sectional view of the first tubing **2106** is shown, according to an exemplary embodiment. In the embodiment shown, the first tubing **2106** includes an inner lumen **2400** and four outer lumens **2402**. That is, the first tubing **2106** is shown to include five separate lumens (e.g., channels, bores, pathways) through which air, fluid, and/or other debris may flow. Preferably, fluid and debris flows primarily through the inner lumen **2400**, while air flows through the outer lumens **2402**. At or near the canister **2104**, the path of the outer lumens **2402** is separated from the path of the inner lumen **2400** as shown in FIG. **18**. The inner lumen **2400** is connected to the inner volume of the canister to allow fluid and debris from the wound space **2120** to be collected in the canister. The outer lumens **2402** are connected to the sensor **2124** to facilitate the measurement and monitoring of pressure at the wound space.

(178) The connection pad **2121** may include grooves and other physical features configured to direct fluid and debris towards the inner lumen **2400** and away from the outer lumens **2402**. However, fluid and debris may occasionally reach one or more of the outer lumens **2402** and cause a blockage of the one or more of the outer lumens **2402**. A blockage of the inner lumen **2400** may also occur. As described below with reference to FIG. **22**, the valve **2126** may be controlled to allow a blast of air to be released through the outer lumens **2402** to clear the fluid or other blockage from the outer lumens **2402**, i.e., by pushing air and fluid back towards the dressing **2102** and out of the outer lumens **2402**.

(179) Referring now to FIG. **22**, a flowchart of a process **2500** for clearing blockages in the first tubing **2106** is shown, according to an exemplary embodiment. At step **2502**, a potential blockage of one or more outer lumens **2402** is determined. As one example, the control circuit **2132** may detect a blockage based on pressure measurements from the sensor **2124**. As another example, the control circuit **2132** assumes a potential blockage exists after a predetermined time period, i.e., such that steps **2504-2508** are triggered at a predetermined frequency.

(180) At step **2504**, the valve **2126** is opened to allow a blast of air therethrough. For example, the control circuit **2132** may provide a non-zero voltage to the solenoid **2206** of the valve **2126**. The control circuit **2132** may cause the valve **2126** to be held in the open position for an extended time period, i.e., longer than the periods shown in FIG. **22** and discussed with reference thereto below. For example, in one embodiment the non-zero voltage is provided for approximately 500

milliseconds to hold the valve **2126** open for approximately 500 milliseconds. When the valve **2126** is held open, a blast of air may flow therethrough at a high airflow rate due to the pressure differential between the surrounding environment (ambient air) and the sealed space. This blast of air may flow into a blocked outer lumen **2402** and push any blockage out of the first tubing **2106** towards the dressing **2106**. Blockages in the outer lumens **2402** may thereby be periodically cleared to allow free airflow through the outer lumens **2402**, for example to ensure that the measurements of the pressure sensor **2124** accurately represent the pressure at the wound space **2120**.

(181) At step **2506**, the solenoid valve **2508** is closed. For example, the control circuit **2132** causes approximately zero voltage to be provided across the solenoid **2206**. Airflow from the environment to the sealed space is prevented. The pneumatic pump **2122** may be operated to reestablish a desired negative pressure at the wound space **2120**.

(182) Referring now to FIG. **23**, a flowchart of a process **2600** for wound volume determination and wound therapy customization is shown, according to an exemplary embodiment. The process **2600** may be carried out by the NPIWT system **100** of FIG. **18**.

(183) At step **2602**, a sealed space defined by the wound **2114**, the dressing **2102**, the first tubing **2106**, and the canister **2104** is established. The sealed space includes the wound space **2120**. In other words, the dressing **2102** is applied to the wound **2114** with the drape **2116** sealed over the wound **2114** and the foam layer **2118** (or other layers included in the dressing **2102** in various embodiments) to define the wound space **2120**. The first tubing **2106** is coupled to the drape **2116** in fluid communication with the wound space **2120** via the connection pad **2121**. The first tubing **2106** is also coupled to the canister **2104** in fluid communication with the canister **2104**.

(184) At step **2604**, the pneumatic pump **2122** is operated to draw a negative pressure in the sealed space. That is, the control circuit **2132** provides a control signal to the pneumatic pump **2122** that causes the pneumatic pump to remove air from the sealed space. The control circuit **2132** may receive pressure measurements from pressure sensor **2124** and cause the pneumatic pump **2122** to cease operation when a desired negative pressure is achieved (e.g., -125 mmHg) and/or otherwise control the pneumatic pump **2122** based on the pressure measurements to provide a desired negative pressure or pattern of desired negative pressures.

(185) At step **2606**, the valve **2126** is repeatedly opened and closed (e.g., “cycled”) to allow a controlled rate of airflow therethrough. The control circuit **2132** may provide a control signal to the valve **2126** that causes the valve **2126** to repeatedly open and close. For example, in an embodiment where the valve **2126** is a solenoid valve, for example as shown in FIGS. **19-20**, at step **2606** the control circuit **2132** provides a voltage pattern to the valve **2126**. That is, the control circuit **2132** may repeatedly alternate a voltage differential across between the positive lead **2210** and the negative lead **2212** between approximately zero volts and a non-zero voltage (e.g., approximately five volts). For example, the voltage pattern may include a step function that repeatedly steps between approximately zero voltage and the non-zero voltage.

(186) In one example, as shown in FIG. **24** and described in detail with reference thereto, the voltage pattern may include a repeating pattern of approximately 400 milliseconds at the non-zero voltage, approximately 100 milliseconds at approximately zero voltage, approximately 400 milliseconds at the non-zero voltage, and approximately 100 milliseconds at approximately zero voltage. The voltage pattern may thereby cause the valve **2126** to alternate between the open position and the closed position with a period of approximately 500 milliseconds. In some embodiments, the voltage pattern may include a first set of two repetitions of the repeating pattern, followed by approximately one second at approximately zero voltage, followed by a second set of two repetitions of the repeating pattern. In preferred embodiments, the non-zero voltage is provided, in each repetition, for no more than a maximum continuous duration of approximately 500 milliseconds.

(187) By controlling the valve **2126** to repeatedly alternate between the open position and the closed position, a controlled rate of airflow is allowed therethrough across the repetitions. That is, a

lower rate of airflow is allow through the valve **2126** as compared to holding the valve **2126** open for an extended or indefinite amount of time (e.g., as described for process **500**), for example 500 milliseconds or longer. The controlled rate may be customized by altering the voltage pattern. Additionally, the controlled rate may be known based on the voltage pattern. For example, the controlled rate may be predetermined by bench testing for each of one or more voltage patterns. In preferred embodiments, the controlled rate is less than a restriction rate of the filter **2138**.

(188) At step **608**, the pressure in the sealed space is measured as the negative pressure in the sealed space decays towards ambient pressure (i.e., approaches approximately atmospheric pressure). The controlled airflow through the valve **2126** allows air to enter the sealed space and causes the pressure in the sealed space to decay towards ambient pressure. The sensor **2124** may measure the pressure in the sealed space and provide the pressure measurements to the control circuit **2132**. The control circuit **2132** may record (store, save) the pressure measurements. In some embodiments, the control circuit **2132** may collect the pressure measurements to form a pressure decay curve.

(189) At step **610**, the volume of the wound space **2120** is determined based on the pressure measurements. For example, based on the known controlled rate of airflow through the valve **2126** and the measured pressure decay curve, the volume of the sealed space may be determined. The volume of the wound space may then be determined by removing a volume of the canister and tube from the total volume of the sealed space. In some cases, one or more additional valves, sensors, etc. are included to facilitate generation and collection of data for use in wound size determination. Various methods for calculating wound size are possible in various embodiments, for example as described with reference to FIGS. **1-17**.

(190) At step **2612**, the wound size (e.g., the volume of the wound space **2120**) and/or a message relating thereto is displayed on the user interface **2128**. For example, the control circuit **2130** may cause a graphical user interface that includes the wound size to be displayed on a screen of the user interface **2128**. As another example, the control circuit **2130** may determine one or more warnings, progress reports, or other wound-related message based on the wound size and control the user interface **2128** to display the warning, report, or other message. For example, the user interface **2128** may display a graphical representation of change in the volume of the wound space over time.

(191) At step **2614**, a wound therapy is customized based on the volume of the wound space. In some embodiments, the control circuit **2130** automatically customizes a wound therapy based on the determined volume of the wound space **2120**. In other embodiments, a user is facilitated in customizing a wound therapy based on the volume of the wound space **2120** based on information displayed on the user interface **2128**.

(192) In the example shown, the control circuit **2130** automatically customizes instillation by automatically determining an amount of instillation fluid to be supplied to the wound space **2120** based on the determined volume of the wound space **2120**. For example, the control circuit **2130** may multiple the determined volume of the wound space **2120** by a scaling factor to determine the amount of instillation fluid to be supplied to the wound space **2120**. As another example, the control circuit **2130** may determine the amount of instillation to be supplied as equal to the volume of the wound space **2120**. Various calculations are possible for various applications, wound types, instillation fluid types, patient and/or caregiver preferences, etc.

(193) At step **2616**, the customized wound therapy is provided. For example, the control circuit **2130** may control the instillation pump **2130** to provide the determined amount of instillation fluid from the instillation fluid source **2110** to the wound space **2120**. Instillation therapy may thereby be tailored to meet the needs of the healing wound in real time. Various other customized therapies are possible in various embodiments.

(194) Referring now to FIG. **24**, a collection of graphs illustrating the operation of the NPIWT system **2100** is shown, according to an exemplary embodiment. FIG. **24** shows a measured pressure graph **2700**, a control signal graph **2702**, and an introduced pressure graph **2704**. The pressure

graph **2700** illustrates a change in pressure in the sealed space over time as measured by the pressure sensor **2124**. As illustrated by the pressure graph **2700**, a measured pressure line **2706** approaches a desired negative pressure (shown as -200 mmHg) as the pneumatic pump **2122** is operated to draw air out of the sealed space. The measured pressure then decays as the valve **2126** is controlled to allow a controlled rate of airflow therethrough.

(195) The control signal graph **2702** illustrates a voltage pattern applied to the valve **2126** (i.e., across the solenoid **2206**). As shown, a control signal **2708** alternates between approximately zero voltage and a non-zero voltage, shown as approximately five volts. As shown, the control signal includes approximately 400 milliseconds at the non-zero voltage, approximately 100 milliseconds at approximately zero voltage, another approximately 400 milliseconds at the non-zero voltage, and another approximately 100 milliseconds at approximately zero voltage. After these two repetitions (i.e., after two periods of 500 milliseconds), the control signal may include one second at approximately zero voltage as illustrated in FIG. **24**. It should be understood that various other frequencies and periods for a voltage pattern may be used in various embodiments. As one possible additional example, in an alternative embodiment the voltage pattern alternates between approximately 200 milliseconds at a non-zero voltage and 50 milliseconds at approximately zero voltage for three or more repetitions (e.g., four repetitions), followed by approximately one second at approximately zero voltage before repeating the voltage pattern. In various embodiments, the non-zero voltage is repeatedly provided for a duration between a minimum continuous duration of approximately 50 milliseconds and a maximum continuous duration of approximately 500 milliseconds, with alternating periods of approximately zero voltage.

(196) The introduced pressure graph **2704** illustrates the amount of pressure let into the sealed space over time. In the example shown, approximately 5 mmHg is introduced into the sealed space for each 400 millisecond segment of non-zero voltage in the control signal **708**. The introduced pressure graph **2704** illustrates that the pressure decay in the sealed space may be managed by the alternating pattern of the valve **2126** (i.e., of the control signal **2708**). For example, the introduced pressure graph **2704** indicates that a lag time may exist between the beginning of a non-zero voltage period and a point in time corresponding to peak rate of pressure reduction or peak rate of airflow through the valve **2126**.

(197) Referring now to FIG. **8**, the NPIWT system **2100** of FIG. **18** is shown in an alternative embodiment. In the embodiment shown, the volume of the sealed space and/or the wound space can be determined by first determining the restriction rate of the filter **2138** (i.e., the rate of airflow through the filter **2138**) as part of a calibration process before therapy is started. As shown in FIG. **18**, a removable cap (structure, cover, joint) **2800** is placed proximate a point wherein the inner lumen **2400** and the outer lumens **2402** come together to form the first tubing **2106** (e.g., at a port of the canister **2104**). For example, the first tubing **2106** may be disconnected at this point and replaced by the removable cap **2800** as shown in FIG. **8**. The removable cap **2800** connects the inner lumen **2400** and the outer lumens **2402** and causes air to flow directly therebetween (i.e., without passing through the dressing **2102**) during a process for determining the restriction rate of the filter **2138**. The removable cap **2800** may then be removed and the first tubing **2106** connected in the configuration described above.

(198) To determine the restriction rate of the filter **2138** while the removable cap **2800** is applied as in FIG. **8**, the valve **2126** is closed and the pneumatic pump **2122** is run to remove air from the canister **2104**. The valve **2126** may then be opened for an indefinite amount of time to allow air to flow back into the canister **2104** while the pressure sensor **2124** measures the change in pressure over time. Based on a known volume of the canister **2104** and the change in pressure over time while the valve **2126** is open, the control circuit **2132** may calculate the rate of airflow through the filter. The cap **2800** facilitates this process by ensuring that the unknown volume of the wound space does not influence the rate of change of the pressure during such a process.

Configuration of Exemplary Embodiments

(199) As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(200) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(201) The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(202) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure. Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, calculation steps, processing steps, comparison steps, and decision steps.

(203) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

(204) As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be

embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

(205) The “circuit” may also include one or more processors communicably coupled to one or more memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc. In some embodiments, the one or more processors may be external to the apparatus, for example the one or more processors may be a remote processor (e.g., a cloud based processor). Alternatively or additionally, the one or more processors may be internal and/or local to the apparatus. In this regard, a given circuit or components thereof may be disposed locally (e.g., as part of a local server, a local computing system, etc.) or remotely (e.g., as part of a remote server such as a cloud based server). To that end, a “circuit” as described herein may include components that are distributed across one or more locations. The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Claims

1. A control circuit for a negative pressure therapy system, the control circuit communicatively coupled to a pump that is fluidly communicable with a sealed space, a sensor fluidly coupled to the sealed space, and a valve fluidly coupled between a sealed space and a surrounding environment, the control circuit configured to: actuate the pump to draw fluid from the sealed space, repeatedly actuate the valve between an open position and a closed position, allowing a controlled rate of airflow through the valve; receive measurements of a pressure in the sealed space generated by the sensor during a pressure decay in response to the controlled rate of airflow through the valve; and in response to the measurements of the pressure and the controlled rate of airflow, determine a volume of the sealed space.
 2. The control circuit of claim 1, wherein the controlled rate of airflow is less than a restriction rate of a filter fluidly coupled between the valve and a canister.
 3. The control circuit of claim 1, wherein the valve comprises a solenoid valve and the control circuit is further configured to actuate the valve by providing a voltage pattern to the solenoid valve.
 4. The control circuit of claim 3, wherein the voltage pattern comprises a step function between a zero voltage and a non-zero voltage.
 5. The control circuit of claim 4, wherein the voltage pattern remains at the non-zero voltage for no more than a maximum continuous duration of 500 milliseconds.
 6. The control circuit of claim 5, wherein voltage pattern comprises a repeating pattern of 400 milliseconds at a non-zero voltage, 100 milliseconds at zero voltage, 400 milliseconds at the non-zero voltage, and 100 milliseconds at zero voltage.
 7. The control circuit of claim 6, wherein the voltage pattern comprises a first set of two periods of the repeating pattern, one second at zero voltage, and a second set of two periods of the repeating pattern.
 8. The control circuit of claim 3, wherein the voltage pattern causes the solenoid valve to alternate between the open position and the closed position with a period of 500 milliseconds.
 9. The control circuit of claim 1, wherein the control circuit is further configured to determine a customized wound therapy based on the volume of the sealed space and to provide the customized wound therapy.
 10. The control circuit of claim 9, wherein the customized wound therapy comprises instillation therapy.
 11. The control circuit of claim 10, wherein the control circuit is configured to customize the instillation therapy by determining an amount of instillation fluid to supply to the sealed space based on the volume of the sealed space.
 12. The control circuit of claim 11, wherein the control circuit is further configured to actuate an instillation pump to provide the amount of the instillation fluid to the sealed space.
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